Part 2 Exp, Log and Trig Integrals - Homework Answers

Revision | Warm-up

1 Exercise 6B Q1

a Here $a = 3$ and $b = 0$, so $y' = 4 \times 3e^{3x} = 12e^{3x}$.

b Here $a = \frac{1}{3}$ and $b = 0$, so $y' = 6 \times \frac{1}{3}e^{\frac{1}{3}x} = 2e^{\frac{1}{3}x}$.

c Here $a = -2$ and $b = 0$, so $y' = -\frac{1}{2} \times (-2)e^{-2x} = e^{-2x}$.

d Here $a = -3$ and $b = 4$, so $y' = -3e^{-3x+4}$.

e Here $a = -2$ and $b = -7$, so $y' = -2e^{-2x-7}$.

2 Exercise 6G Q8

a Let $u = x^2 + 3x + 2$. Then $u' = 2x + 3$, so

$$\frac{d}{dx}\log_e(x^2 + 3x + 2) = \frac{2x + 3}{x^2 + 3x + 2}.$$

b Let $u = 1 + 2x^3$. Then $u' = 6x^2$, so $\frac{d}{dx}\log_e(1 + 2x^3) = \frac{6x^2}{1 + 2x^3}$.

c Let $u = e^x - 2$. Then $u' = e^x$, so $\frac{d}{dx}\ln(e^x - 2) = \frac{e^x}{e^x - 2}$.

3 Exercise 6G Q5

a $y = \ln x^3 = 3\ln x$ (the log of a power), so $y' = 3 \times \frac{1}{x} = \frac{3}{x}$.

b $y = \ln x^{-3} = -3\ln x$, so $y' = -3 \times \frac{1}{x} = -\frac{3}{x}$.

c $y = \ln x^{\frac{1}{2}} = \frac{1}{2}\ln x$, so $y' = \frac{1}{2} \times \frac{1}{x} = \frac{1}{2x}$.

d $y = \ln(x+1)^{\frac{1}{2}} = \frac{1}{2}\ln(x+1)$, so $y' = \frac{1}{2} \times \frac{1}{x+1} = \frac{1}{2(x+1)}$.

4 Exercise 7B Q2

a Here $a = 2$, so $\frac{d}{dx}\sin(2x - 1) = 2\cos(2x - 1)$.

b Here $a = -1$, so $\frac{d}{dx}2\cos(1 - x) = 2 \times \big(-(-1)\sin(1 - x)\big) = 2\sin(1 - x)$.

c Here $a = -3$, so

$$\frac{d}{dx}7\sin(2 - 3x) = 7 \times (-3)\cos(2 - 3x) = -21\cos(2 - 3x).$$

d Write $\frac{x+1}{2} = \frac{1}{2}x + \frac{1}{2}$, so $a = \frac{1}{2}$ and

$$\frac{d}{dx}6\sin\left(\frac{x+1}{2}\right) = 6 \times \tfrac{1}{2}\cos\left(\frac{x+1}{2}\right) = 3\cos\left(\frac{x+1}{2}\right).$$

5 Exercise 7B Q4

$f(x) = \cos 2x$, so $f'(x) = -2\sin 2x$.

a $f'(0) = -2\sin 0 = -2 \times 0 = 0$

b $f'\left(\frac{\pi}{12}\right) = -2\sin\frac{\pi}{6} = -2 \times \frac{1}{2} = -1$

c $f'\left(\frac{\pi}{6}\right) = -2\sin\frac{\pi}{3} = -2 \times \frac{\sqrt{3}}{2} = -\sqrt{3}$

d $f'\left(\frac{\pi}{4}\right) = -2\sin\frac{\pi}{2} = -2 \times 1 = -2$

6 Exercise 5B Q4

a
$$\begin{aligned}\int_{-2}^{1} 3x^2\,dx &= \Big[x^3\Big]_{-2}^{1} \\ &= 1^3 - (-2)^3 \\ &= 1 - (-8) \\ &= 9\end{aligned}$$

b
$$\begin{aligned}\int_{-1}^{2} (4x^3 + 5)\,dx &= \Big[x^4 + 5x\Big]_{-1}^{2} \\ &= (16 + 10) - (1 - 5) \\ &= 26 - (-4) \\ &= 30\end{aligned}$$

c
$$\begin{aligned}\int_{-2}^{2} (5x^4 + 6x^2)\,dx &= \Big[x^5 + 2x^3\Big]_{-2}^{2} \\ &= (32 + 16) - (-32 - 16) \\ &= 48 - (-48) \\ &= 96\end{aligned}$$

The integrand is even, so this is also $2\int_{0}^{2} (5x^4 + 6x^2)\,dx = 2 \times 48 = 96$.

d
$$\begin{aligned}\int_{-2}^{-1} (4x^3 + 12x^2 - 3)\,dx &= \Big[x^4 + 4x^3 - 3x\Big]_{-2}^{-1} \\ &= (1 - 4 + 3) - (16 - 32 + 6) \\ &= 0 - (-10) \\ &= 10\end{aligned}$$

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 6D | Integration of exponential functions

Lesson Examples

C1

Tutor demo. Use $\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C$.

a Here $a = 5$ and $b = -1$, so $\int e^{5x-1}\,dx = \frac{1}{5}e^{5x-1} + C$.

b Integrating each term separately, $\int (2 + 3x - e^x)\,dx = 2x + \frac{3}{2}x^2 - e^x + C$.

c Here $a = -2$ and $b = 4$, so $\int e^{4-2x}\,dx = \dfrac{1}{-2}\,e^{4-2x} + C = -\frac{1}{2}e^{4-2x} + C$.

Your turn.

a Here $a = 2$ and $b = 7$, so $\int e^{2x+7}\,dx = \frac{1}{2}e^{2x+7} + C$.

b Integrating each term separately, $\int (4 - x + e^x)\,dx = 4x - \frac{1}{2}x^2 + e^x + C$.

c Here $a = -3$ and $b = 1$, so $\int e^{1-3x}\,dx = \dfrac{1}{-3}\,e^{1-3x} + C = -\frac{1}{3}e^{1-3x} + C$.

C2

Tutor demo.

a

$$\begin{aligned}\int_0^3 e^x\,dx &= \Big[e^x\Big]_0^3 \\ &= e^3 - e^0 \\ &= e^3 - 1\end{aligned}$$

b $(a = -3 \text{ and } b = 7)$

$$\begin{aligned}\int_2^3 e^{7-3x}\,dx &= -\tfrac{1}{3}\Big[e^{7-3x}\Big]_2^3 \\ &= -\tfrac{1}{3}\left(e^{-2} - e^1\right) \\ &= -\frac{1}{3}\left(\frac{1}{e^2} - e\right) \\ &= \frac{e^3 - 1}{3e^2}\end{aligned}$$

In **b**, $7 - 3 \times 3 = -2$ and $7 - 3 \times 2 = 1$, and the last line multiplies top and bottom by e^2.

Your turn.

a

$$\begin{aligned}\int_{-1}^1 e^x\,dx &= \Big[e^x\Big]_{-1}^1 \\ &= e^1 - e^{-1} \\ &= \frac{e^2 - 1}{e}\end{aligned}$$

b $(a = -4 \text{ and } b = 2)$

$$\begin{aligned}\int_0^1 e^{2-4x}\,dx &= -\tfrac{1}{4}\Big[e^{2-4x}\Big]_0^1 \\ &= -\tfrac{1}{4}\left(e^{-2} - e^2\right) \\ &= -\frac{1}{4}\left(\frac{1}{e^2} - e^2\right) \\ &= \frac{e^4 - 1}{4e^2}\end{aligned}$$

C3 Tutor demo.

1a Taking the primitive, $f(x) = e^x + C$.
Substituting $f(2) = 0,\ 0 = e^2 + C$,
so $C = -e^2$.
Hence $f(x) = e^x - e^2$.

1b $f(0) = e^0 - e^2 = 1 - e^2$

2a Taking the primitive,
$y = 2x - 3e^{-x} + C$.
Substituting $y(0) = 1$,
$1 = 0 - 3e^0 + C$,
so $1 = -3 + C$ and $C = 4$.
Hence $y = 2x - 3e^{-x} + 4$.

2b $y(1) = 2 - 3e^{-1} + 4 = 6 - 3e^{-1}$

Your turn.

1a Taking the primitive, $f(x) = e^x + C$.
Substituting $f(-1) = 0,\ 0 = e^{-1} + C$,
so $C = -e^{-1}$.
Hence $f(x) = e^x - e^{-1}$.

1b $f(0) = e^0 - e^{-1} = 1 - e^{-1}$

2a Taking the primitive,
$y = 3x + e^{-x} + C$.
Substituting $y(0) = 2,\ 2 = 0 + e^0 + C$,
so $2 = 1 + C$ and $C = 1$.
Hence $y = 3x + e^{-x} + 1$.

2b $y(1) = 3 + e^{-1} + 1 = 4 + e^{-1}$

C4 Tutor demo.

a Let $u = x^4$, so $y = e^u$, $\dfrac{du}{dx} = 4x^3$ and $\dfrac{dy}{du} = e^u$. Then
$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx} = 4x^3e^{x^4}$.

b Reversing part **a**, $\displaystyle\int 4x^3e^{x^4}\,dx = e^{x^4}$, so
$\displaystyle\int_{-1}^{1} 4x^3e^{x^4}\,dx = \left[e^{x^4}\right]_{-1}^{1} = e^1 - e^1 = 0$, because $(-1)^4 = 1$.

c The function $4x^3e^{x^4}$ is odd, because $4(-x)^3e^{(-x)^4} = -4x^3e^{x^4}$, and the interval $[-1, 1]$ is symmetric in the y-axis, so the integral is zero.

Your turn.

a Let $u = 2x^2$, so $y = e^u$, $\dfrac{du}{dx} = 4x$ and $\dfrac{dy}{du} = e^u$. Then
$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx} = 4xe^{2x^2}$.

b Reversing part **a**, $\displaystyle\int 4xe^{2x^2}\,dx = e^{2x^2}$, so
$\displaystyle\int_{-1}^{1} 4xe^{2x^2}\,dx = \left[e^{2x^2}\right]_{-1}^{1} = e^2 - e^2 = 0$, because $2(-1)^2 = 2$.

c The function $4xe^{2x^2}$ is odd, because $4(-x)e^{2(-x)^2} = -4xe^{2x^2}$, and the interval $[-1, 1]$ is symmetric in the y-axis, so the integral is zero.

C5 **Tutor demo.** Let $u = 1 + e^{2x}$. Then $u' = 2e^{2x}$, and with $n = -3$,

$$\int u^{-3}\,\frac{du}{dx}\,dx = \frac{u^{-2}}{-2}.$$

$$\begin{aligned}\int \frac{e^{2x}}{(1+e^{2x})^3}\,dx &= \tfrac{1}{2}\int \frac{2e^{2x}}{(1+e^{2x})^3}\,dx \\ &= \tfrac{1}{2}\times\left(-\tfrac{1}{2}\right)\times(1+e^{2x})^{-2} \\ &= -\frac{1}{4(1+e^{2x})^2}+C\end{aligned}$$

Your turn. Let $u = 1 - e^{3x}$. Then $u' = -3e^{3x}$, and with $n = -2$,

$$\int u^{-2}\,\frac{du}{dx}\,dx = \frac{u^{-1}}{-1}.$$

$$\begin{aligned}\int \frac{e^{3x}}{(1-e^{3x})^2}\,dx &= -\tfrac{1}{3}\int \frac{-3e^{3x}}{(1-e^{3x})^2}\,dx \\ &= -\tfrac{1}{3}\times(-1)\times(1-e^{3x})^{-1} \\ &= \frac{1}{3(1-e^{3x})}+C\end{aligned}$$

Textbook Homework

1 a $\frac{1}{2}e^{2x}+C$ b $\frac{1}{3}e^{3x}+C$ c $3e^{\frac{1}{3}x}+C$ d $2e^{\frac{1}{2}x}+C$ e $5e^{2x}+C$
f $4e^{3x}+C$ g $\frac{1}{4}e^{4x+5}+C$ h $\frac{1}{4}e^{4x-2}+C$ i $2e^{3x+2}+C$ j $e^{4x+3}+C$
k $-\frac{1}{2}e^{7-2x}+C$ l $-\frac{1}{6}e^{1-3x}+C$

2 a $e-1$ b e^2-e c $e-e^{-3}$ d e^2-1 e $\frac{1}{2}(e^4-1)$ f $4(e^5-e^{-10})$
g $2(e^{12}-e^{-4})$ h $\frac{3}{2}(e^{18}-e^{-6})$ i $\frac{1}{2}(e^3-e^{-1})$ j $\frac{1}{4}(e^{-3}-e^{-11})$
k $\frac{1}{3}(e^{-1}-e^{-4})$ l $\frac{e^2}{2}(e^2-1)$ m $\frac{e}{3}(e^2-1)$ n $2e^4(e^3-1)$
o $3e^3(e^4-1)$ p $4e^2(e^3-1)$

3 a $-e^{-x}+C$ b $-\frac{1}{2}e^{-2x}+C$ c $-\frac{1}{3}e^{-3x}+C$ d $e^{-3x}+C$
e $-3e^{-2x}+C$ f $4e^{2x}+C$

4 a $f(x)=\frac{1}{2}e^{2x}+C$, for some constant C
b $C=-\frac{5}{2}$, so $f(x)=\frac{1}{2}e^{2x}-\frac{5}{2}$
c $f(1)=\frac{1}{2}e^2-\frac{5}{2}$, $f(2)=\frac{1}{2}e^4-\frac{5}{2}$

5 a $f(x)=x+2e^x-1$, $f(1)=2e$ b $f(x)=2+x-3e^x$, $f(1)=3-3e$
c $f(x)=1+2x-e^{-x}$, $f(1)=3-e^{-1}$
d $f(x)=1+4x+e^{-x}$, $f(1)=5+e^{-1}$
e $f(x)=\frac{1}{2}e^{2x-1}+\frac{5}{2}$, $f(1)=\frac{1}{2}(e+5)$
f $f(x)=1-\frac{1}{3}e^{1-3x}$, $f(1)=1-\frac{1}{3}e^{-2}$
g $f(x)=2e^{\frac{1}{2}x+1}-6$, $f(1)=2e^{\frac{3}{2}}-6$
h $f(x)=3e^{\frac{1}{3}x+2}-1$, $f(1)=3e^{\frac{7}{3}}-1$

6 a $\frac{1}{2}e^{2x}+e^x+C$ b $\frac{1}{2}e^{2x}-e^x+C$ c $e^{-x}-e^{-2x}+C$
d $\frac{1}{2}e^{2x}+2e^x+x+C$ e $\frac{1}{2}e^{2x}-2e^x+x+C$ f $\frac{1}{2}e^{2x}-4e^x+4x+C$
g $\frac{1}{2}(e^{2x}+e^{-2x})+C$ h $\frac{1}{10}(e^{10x}+e^{-10x})+C$

7 **a** $\frac{1}{7}e^{7x+q}+C$ **b** $\frac{1}{3}e^{3x-k}+C$ **c** $\frac{1}{s}e^{sx+1}+C$ **d** $\frac{1}{k}e^{kx-1}+C$
e $e^{px+q}+C$ **f** $e^{mx+k}+C$ **g** $\frac{A}{s}e^{sx-t}+C$

8 **a** $-e^{1-x}+C$ **b** $-\frac{1}{3}e^{1-3x}+C$ **c** $-\frac{1}{2}e^{-2x-5}+C$ **d** $-2e^{1-2x}+C$
e $2e^{5x-2}+C$ **f** $-4e^{5-3x}+C$

9 **a** $x-e^{-x}+C$ **b** $e^{x}-e^{-x}+C$ **c** $\frac{1}{2}e^{-2x}-e^{-x}+C$
d $e^{-3x}-\frac{1}{2}e^{-2x}+C$ **e** $e^{-3x}-e^{-2x}+C$ **f** $e^{-x}-e^{-2x}+C$

10 **a** $y=e^{x-1}$, y-intercept e^{-1}
b $y=e^{2}+1-e^{2-x}$, horizontal asymptote $y=e^{2}+1$
c $f(x)=e^{x}+\frac{x}{e}-1$, $f(0)=0$
d $f(x)=e^{x}-e^{-x}-2x$

11 **a** $e^{2}-e$ **b** $\frac{1}{2}(e^{2}-e^{-2})+4(e-e^{-1})+8$ **c** $e+e^{-1}-2$
d $\frac{1}{4}(e^{4}-e^{-4})+\frac{1}{2}(e^{-2}-e^{2})$ **e** $e-e^{-1}$ **f** $e-e^{-1}+\frac{1}{2}(e^{-2}-e^{2})$

12 **a** (i) $2xe^{x^2+3}$ (ii) $e^{x^2+3}+C$ **b** (i) $2(x-1)e^{x^2-2x+3}$ (ii) $\frac{1}{2}e^{x^2-2x+3}+C$
c (i) $(6x+4)e^{3x^2+4x+1}$ (ii) $\frac{1}{2}e^{3x^2+4x+1}+C$ **d** (i) $3x^2e^{x^3}$ (ii) $\frac{1}{3}(1-e^{-1})$

13 **a** $-\frac{1}{2}e^{-2x}+C$ **b** $-\frac{1}{3}e^{-3x}+C$ **c** $2e^{\frac{1}{2}x}+C$ **d** $3e^{\frac{1}{3}x}+C$
e $-2e^{-\frac{1}{2}x}+C$ **f** $-3e^{-\frac{1}{3}x}+C$

14 **a** $y'=xe^{x}+e^{x}$, $e^{2}+1$ **b** $y'=-xe^{-x}+e^{-x}$, $-1-e^{2}$

15 **a** $2e^{\frac{1}{2}x}+\frac{2}{3}e^{-\frac{3}{2}x}+C$ **b** $\frac{3}{2}e^{\frac{2}{3}x}-\frac{3}{4}e^{-\frac{4}{3}x}+C$

16 **b** 0

17 **a** $\frac{1}{2}e^{x^2}+C$ **b** $2e^{x^2-7}+C$ **c** $\frac{1}{2}e^{3x^2+4x+1}+C$ **d** $\frac{1}{3}e^{x^3-3x^2}+C$
e $-e^{x^{-1}}+C$ **f** $\frac{2}{3}e^{-x\sqrt{x}}+C$

20 **b** $\alpha\doteq 1.1276$ **d** $e^{0.5}=\alpha+\sqrt{\alpha^2-1}\doteq 1.65$, $e^{-0.5}=\alpha-\sqrt{\alpha^2-1}\doteq 0.61$
(the calculator gives $e^{0.5}=1.6487\ldots$ and $e^{-0.5}=0.6065\ldots$, which agree)

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 6D | Integration of exponential functions

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Boys High School, 2020, Year 12 Term 3 Trial HSC, Mathematics Extension 1, Q11(b)**

(i) Using the product rule, let $y = uv$, where $u = x$ and $v = e^{-x}$. Then $u' = 1$ and $v' = -e^{-x}$, so

$$\begin{aligned}\frac{dy}{dx} &= u'v + v'u \\ &= 1 \times e^{-x} + (-e^{-x}) \times x \\ &= e^{-x} - xe^{-x} \\ &= e^{-x}(1-x).\end{aligned}$$

(ii) Reversing part (i),

$$\int_0^1 \left(e^{-x} - xe^{-x}\right) dx = \Big[xe^{-x}\Big]_0^1.$$

Split the left-hand side into two integrals and make the required integral the subject:

$$\begin{aligned}\int_0^1 xe^{-x}\,dx &= \int_0^1 e^{-x}\,dx - \Big[xe^{-x}\Big]_0^1 \\ &= \Big[-e^{-x}\Big]_0^1 - \Big[xe^{-x}\Big]_0^1 \\ &= \left(-e^{-1} - (-e^0)\right) - \left(1 \times e^{-1} - 0 \times e^0\right) \\ &= -e^{-1} + 1 - e^{-1} \\ &= 1 - \frac{2}{e}.\end{aligned}$$

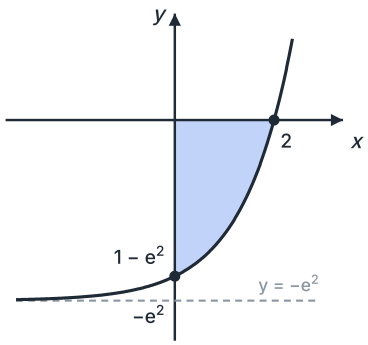
Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 6E | Applications of integration

Lesson Examples

C1 **Tutor demo.**

a Move the graph of $y = e^x$ down e^2 units.
When $x = 0$, $y = e^0 - e^2 = 1 - e^2$.
When $y = 0$, $e^x = e^2$, so $x = 2$.
The horizontal asymptote moves down to $y = -e^2$.



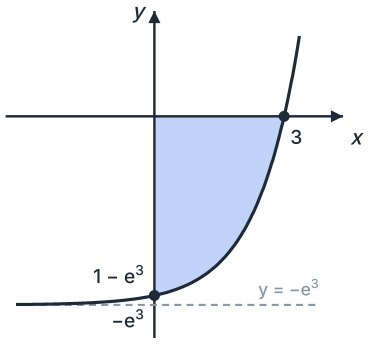
b The region is below the x-axis, from $x = 0$ to $x = 2$.

$$\begin{aligned}\int_0^2 (e^x - e^2)\,dx &= \left[e^x - e^2 x\right]_0^2 \\ &= (e^2 - 2e^2) - (e^0 - 0) \\ &= -e^2 - 1.\end{aligned}$$

The integral is negative because the region is below the x-axis. Hence the area is $e^2 + 1$ square units.

Your turn.

a Move the graph of $y = e^x$ down e^3 units.
When $x = 0$, $y = e^0 - e^3 = 1 - e^3$.
When $y = 0$, $e^x = e^3$, so $x = 3$.
The horizontal asymptote moves down to $y = -e^3$.



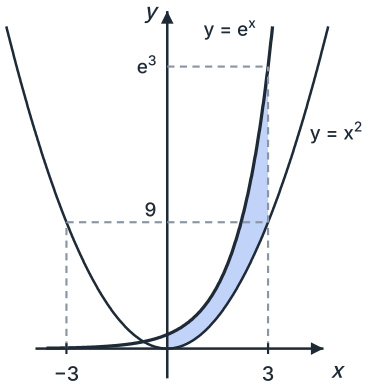
b The region is below the x-axis, from $x = 0$ to $x = 3$.

$$\begin{aligned}\int_0^3 (e^x - e^3)\,dx &= \left[e^x - e^3 x\right]_0^3 \\ &= (e^3 - 3e^3) - (e^0 - 0) \\ &= -2e^3 - 1.\end{aligned}$$

The integral is negative because the region is below the x-axis. Hence the area is $2e^3 + 1$ square units.

C2 **Tutor demo.**

a The graphs are drawn to the right. For $x > 0$, $y = e^x$ is always above $y = x^2$. At $x = 3$, $e^3 \doteq 20.1$ and $3^2 = 9$.



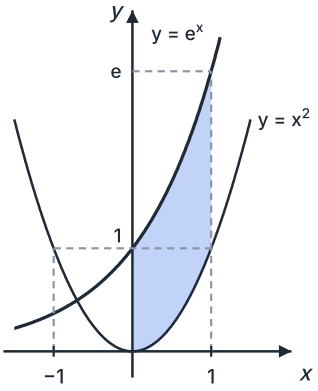
b Using the formula, with e^x on top,

$$\begin{aligned}\text{area} &= \int_0^3 \left(e^x - x^2\right) dx \\ &= \left[e^x - \tfrac{1}{3}x^3\right]_0^3 \\ &= \left(e^3 - 9\right) - \left(e^0 - 0\right) \\ &= e^3 - 10 \text{ square units.}\end{aligned}$$

Here $\frac{1}{3} \times 3^3 = 9$ and $e^0 = 1$.

Your turn.

a The graphs are drawn to the right. For $x > 0$, $y = e^x$ is always above $y = x^2$.



b Using the formula, with e^x on top,

$$\begin{aligned}\text{area} &= \int_0^1 \left(e^x - x^2\right) dx \\ &= \left[e^x - \tfrac{1}{3}x^3\right]_0^1 \\ &= \left(e - \tfrac{1}{3}\right) - \left(e^0 - 0\right) \\ &= e - 1\tfrac{1}{3} \text{ square units.}\end{aligned}$$

Textbook Homework

1 **a i** $e - 1 \doteq 1.72$ **ii** $1 - e^{-1} \doteq 0.63$ **iii** $1 - e^{-2} \doteq 0.86$ **iv** $1 - e^{-3} \doteq 0.95$

b About 172 little squares, so about 1.72, which agrees with part **a**.

c About 63, 86 and 95 little squares, so about 0.63, 0.86 and 0.95, which agree with part **a**.

d About 100 little squares, so about 1. (The total area is exactly 1.)

2 **a** $\frac{1}{2}(e^6 - 1) \doteq 201.2$ square units **b** $1 - e^{-1} \doteq 0.6321$ square units
c $3(1 - e^{-1}) \doteq 1.896$ square units

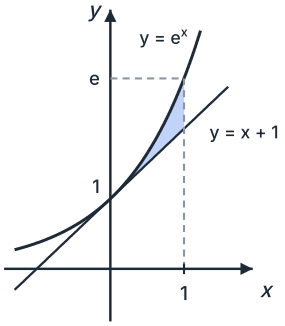
3 **a** $e(e^2 - 1)$ u^2 **b** $\frac{1}{2}(e - e^{-1})$ u^2 **c** $\frac{1}{2}e(e^2 - 1)$ u^2 **d** $3e^2(e - 1)$ u^2

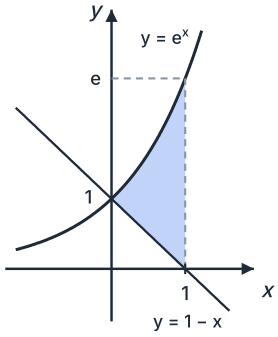
4 **a** $2(e - e^{-\frac{1}{2}})$ u^2 **b** $(1 - e^{-1})$ u^2

5 **a** $2(e^2 - e^{-2}) \doteq 14.51$ u^2 **b** $18 + e^3 - e^{-3} \doteq 38.04$ u^2

6 **a** $(1+e^{-2})$ u^2 **b** 1 u^2 **c** e^{-1} u^2 **d** $(3+e^{-2})$ u^2 **e** 1 u^2
f $(9+e^{-2}-e)$ u^2

7 **a** $\int_0^1 (e^x - 1 - x)\,dx = \left(e - 2\frac{1}{2}\right)$ u^2 **b** $\int_0^1 (e^x - 1 + x)\,dx = \left(e - 1\frac{1}{2}\right)$ u^2



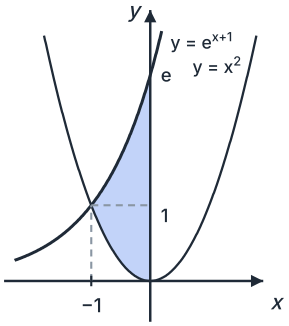


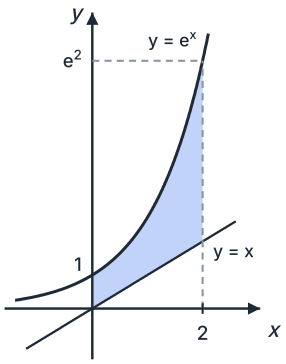
8 **a** The region is symmetric in the y-axis, so the area is twice the area in the first quadrant, where the region is below $y = e^{-x}$.
b $2 - \dfrac{2}{e}$ square units

9 **a** The region is symmetric in the y-axis, so the area is twice the area in the first quadrant, where the region is below $y = e - e^x$.
b 2 square units

10 **b** 0
c The region is symmetric under a half-turn about the origin, so the area is twice the area of the part in the first quadrant.
d $2(e^3 + e^{-3} - 2)$ square units

11 **b** **c** $\left(e - 1\frac{1}{3}\right)$ u^2



12 

$(e^2 - 3)$ square units

13 **a** $e - 1 \doteq 1.7183$ **b** 1.7539

c The trapezoidal-rule approximation is greater. The curve is concave up, so all the chords are above the curve.

14 **a** $\dfrac{d}{dx}e^{-x^2} = -2xe^{-x^2}$, so a primitive of xe^{-x^2} is $-\frac{1}{2}e^{-x^2}$.

b From $x = 0$ to $x = 2$, area $= \frac{1}{2} - \frac{1}{2}e^{-4}$ square units. The function is odd, so the area (not signed) from $x = -2$ to $x = 2$ is $1 - e^{-4}$ square units.

15 **a** (i) $1 - e^{N}$ (ii) 1 **b** (i) $1 - e^{-N}$ (ii) 1

c $\displaystyle\int_0^N 2xe^{-x^2}\,dx = 1 - e^{-N^2}$, thus in the limit as $N \to \infty$ this is just 1.

16 **a** $2\left(e - e^{\sqrt{\delta}}\right)$ **b** It approaches $2(e - 1)$.

17 **a** $\dfrac{d}{dx}\left(xe^{-x}\right) = e^{-x} - xe^{-x}$, $\displaystyle\int_0^N xe^{-x}\,dx = 1 - (1 + N)e^{-N}$ **b** 1

c $\dfrac{d}{dx}\left(x^2e^{-x}\right) = 2xe^{-x} - x^2e^{-x}$, $\displaystyle\int_0^\infty x^2e^{-x}\,dx = 2$

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 6E | Applications of integration

Past-Paper Homework - Worked Solutions

PP 1 **Pymble Ladies' College, 2023, Mathematics Extension 1 HSC Trial Examination Term 3, Q13(d)**

(i) When $x = e + 1$, $y = \log_e(e + 1 - 1) = \log_e e = 1$.

(ii) The curve meets the x-axis where $\log_e(x - 1) = 0$, that is, $x - 1 = 1$, so $x = 2$. The shaded region runs from $y = 0$ to $y = 1$, with the curve on the left and the line $x = e + 1$ on the right, so integrate with respect to y.

From $y = \log_e(x - 1)$, $x - 1 = e^y$, so $x = e^y + 1$. Using the right-hand boundary minus the left-hand boundary,

$$\begin{aligned}
\text{area} &= \int_0^1 \Big((e + 1) - (e^y + 1)\Big)\, dy \\
&= \int_0^1 (e - e^y)\, dy \\
&= \Big[ey - e^y\Big]_0^1 \\
&= (e \times 1 - e^1) - (e \times 0 - e^0) \\
&= 0 - (-1) \\
&= 1.
\end{aligned}$$

The shaded area is 1 square unit.

PP 2 **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(c)**

(i) When $x = 0$, $f(0) = e^0 + 1 = 2$, so the graph crosses the y-axis at $(0, 2)$.

Differentiating, $f'(x) = \frac{1}{2}e^{\frac{x}{2}}$, so $f'(0) = \frac{1}{2}e^0 = \frac{1}{2}$. The gradient of the tangent at $(0, 2)$ is $\frac{1}{2}$, so the gradient of the normal is -2 (since $\frac{1}{2} \times (-2) = -1$).

$$\begin{aligned}
y - 2 &= -2(x - 0) \\
y &= -2x + 2.
\end{aligned}$$

The normal is $y = -2x + 2$, or $2x + y - 2 = 0$.

(ii) The normal meets the x-axis where $0 = -2x + 2$, that is, at $x = 1$. The shaded region runs from $x = 0$ to $x = 1$ (the vertical edge in the diagram is the line $x = 1$, through the point where the normal meets the x-axis), with the curve on top and the normal below. Top minus bottom:

$$\begin{aligned}\text{area} &= \int_0^1 \left(\left(e^{\frac{x}{2}} + 1\right) - \left(-2x + 2\right)\right) dx \\ &= \int_0^1 \left(e^{\frac{x}{2}} + 2x - 1\right) dx \\ &= \left[2e^{\frac{x}{2}} + x^2 - x\right]_0^1 \\ &= \left(2e^{\frac{1}{2}} + 1 - 1\right) - \left(2e^0 + 0 - 0\right) \\ &= 2\sqrt{e} - 2.\end{aligned}$$

Here $\int e^{\frac{1}{2}x}\, dx = \dfrac{1}{\frac{1}{2}}\, e^{\frac{1}{2}x} = 2e^{\frac{1}{2}x}$. The shaded area is $2\sqrt{e} - 2$ square units.

PP 3 **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(c)(i)(ii)**

(i) Using the product rule with $u = x$ and $v = \ln x$, so $u' = 1$ and $v' = \dfrac{1}{x}$,

$$\begin{aligned}\frac{d}{dx}\left(x \ln x\right) &= u'v + v'u \\ &= 1 \times \ln x + \frac{1}{x} \times x \\ &= \ln x + 1,\end{aligned}$$

which is $1 + \ln x$, as required.

(ii) The inverse function: from $y = 1 + \ln x$, swap x and y to get $x = 1 + \ln y$. Then $\ln y = x - 1$, so $f^{-1}(x) = e^{x-1}$.

The two curves meet on the line $y = x$ at $(1, 1)$, since $f(1) = 1 + \ln 1 = 1$. The curve $y = f(x)$ meets the x-axis where $1 + \ln x = 0$, that is, $x = e^{-1}$. The curve $y = f^{-1}(x)$ meets the y-axis at $y = e^{-1}$.

So the shaded region lies under $y = e^{x-1}$ from $x = 0$ to $x = 1$, with the region under $y = f(x)$ from $x = e^{-1}$ to $x = 1$ taken away:

$$\text{area} = \int_0^1 e^{x-1}\, dx - \int_{e^{-1}}^1 (1 + \ln x)\, dx.$$

By part (i), a primitive of $1 + \ln x$ is $x \ln x$. Hence

$$\begin{aligned}\text{area} &= \left[e^{x-1}\right]_0^1 - \left[x \ln x\right]_{e^{-1}}^1 \\ &= \left(e^0 - e^{-1}\right) - \left(1 \times \ln 1 - e^{-1} \ln e^{-1}\right) \\ &= \left(1 - e^{-1}\right) - \left(0 - e^{-1} \times (-1)\right) \\ &= 1 - e^{-1} - e^{-1} \\ &= 1 - \frac{2}{e}.\end{aligned}$$

The shaded area is $1 - \dfrac{2}{e}$ square units.

PP 4 **Saint Ignatius' College, Riverview, 2024, Mathematics Assessment Task, Year 12 Mathematics (Extension One) Task 4, Trial HSC Examination, Q13(d)(ii)**

(ii) The curve and the line meet at the point of part (i), where $x = \dfrac{3e^2}{2}$ and $y = \ln\left(\dfrac{2}{3} \times \dfrac{3e^2}{2}\right) = \ln e^2 = 2$.

The line meets the x-axis where $0 = -\dfrac{3e^2 x}{2} + \dfrac{9e^4}{4} + 2$, so $\dfrac{3e^2 x}{2} = \dfrac{9e^4}{4} + 2$, and multiplying by $\dfrac{2}{3e^2}$, $x = \dfrac{3e^2}{2} + \dfrac{4}{3e^2}$.

The curve meets the x-axis where $\ln \dfrac{2x}{3} = 0$, that is, $\dfrac{2x}{3} = 1$, so $x = \dfrac{3}{2}$.

Let $O = (0, 0)$, $A = \left(\dfrac{3e^2}{2} + \dfrac{4}{3e^2}, 0\right)$, $B = \left(\dfrac{3e^2}{2}, 2\right)$ and $C = (0, 2)$. The enclosed region is the trapezium $OABC$ with the region between the curve and the y-axis, from $y = 0$ to $y = 2$, taken away.

The trapezium has parallel sides OA and CB and height 2:

$$\begin{aligned}\text{area of } OABC &= \tfrac{1}{2}\left(\frac{3e^2}{2} + \frac{4}{3e^2} + \frac{3e^2}{2}\right) \times 2 \\ &= 3e^2 + \frac{4}{3e^2}.\end{aligned}$$

For the curve, $y = \ln \dfrac{2x}{3}$ gives $e^y = \dfrac{2x}{3}$, so $x = \tfrac{3}{2}e^y$. Integrating with respect to y,

$$\begin{aligned}\int_0^2 \tfrac{3}{2}e^y \, dy &= \tfrac{3}{2}\Big[e^y\Big]_0^2 \\ &= \tfrac{3}{2}\left(e^2 - e^0\right) \\ &= \tfrac{3}{2}e^2 - \tfrac{3}{2}.\end{aligned}$$

Hence the enclosed area is

$$\begin{aligned}3e^2 + \frac{4}{3e^2} - \left(\tfrac{3}{2}e^2 - \tfrac{3}{2}\right) &= 3e^2 - \tfrac{3}{2}e^2 + \tfrac{3}{2} + \frac{4}{3e^2} \\ &= \tfrac{3}{2}e^2 + \tfrac{3}{2} + \frac{4}{3e^2} \text{ square units.}\end{aligned}$$

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 6I | Integration of the reciprocal function

Lesson Examples

C1 **Tutor demo.**

a

$$\begin{aligned}\int_1^3 \frac{1}{x}\,dx &= \Big[\log_e x\Big]_1^3 \\ &= \log_e 3 - \log_e 1 \\ &= \log_e 3\end{aligned}$$

c

$$\begin{aligned}\int_1^{e^3} \frac{1}{x}\,dx &= \Big[\log_e x\Big]_1^{e^3} \\ &= \log_e e^3 - \log_e 1 \\ &= 3 - 0 \\ &= 3\end{aligned}$$

b $\int_1^3 \frac{1}{x}\,dx = \log_e 3 \doteq 1.099$. Both rectangles have width $3 - 1 = 2$. The lower rectangle has height $\frac{1}{3}$, the value of $\frac{1}{x}$ at $x = 3$, so its area is $2 \times \frac{1}{3} = \frac{2}{3}$. The upper rectangle has height 1, the value of $\frac{1}{x}$ at $x = 1$, so its area is $2 \times 1 = 2$. Since $\frac{2}{3} < 1.099 < 2$, the integral is indeed between $\frac{2}{3}$ and 2.

Your turn.

a

$$\begin{aligned}\int_1^5 \frac{1}{x}\,dx &= \Big[\log_e x\Big]_1^5 \\ &= \log_e 5 - \log_e 1 \\ &= \log_e 5\end{aligned}$$

c

$$\begin{aligned}\int_1^{e^4} \frac{1}{x}\,dx &= \Big[\log_e x\Big]_1^{e^4} \\ &= \log_e e^4 - \log_e 1 \\ &= 4 - 0 \\ &= 4\end{aligned}$$

b $\int_1^5 \frac{1}{x}\,dx = \log_e 5 \doteq 1.609$. Both rectangles have width $5 - 1 = 4$. The lower rectangle has height $\frac{1}{5}$, so its area is $4 \times \frac{1}{5} = \frac{4}{5}$. The upper rectangle has height 1, so its area is $4 \times 1 = 4$. Since $\frac{4}{5} < 1.609 < 4$, the integral is indeed between $\frac{4}{5}$ and 4.

C2 Tutor demo.

a

$$\begin{aligned}\int_e^{e^3} \frac{4}{x}\,dx &= 4\Big[\log_e|x|\Big]_e^{e^3} \\ &= 4\big(\log_e e^3 - \log_e e\big) \\ &= 4(3-1) \\ &= 8\end{aligned}$$

b $(a=-3 \text{ and } b=1)$

$$\begin{aligned}\int_1^3 \frac{1}{1-3x}\,dx &= -\tfrac{1}{3}\Big[\log_e|1-3x|\Big]_1^3 \\ &= -\tfrac{1}{3}\big(\log_e|-8| - \log_e|-2|\big) \\ &= -\tfrac{1}{3}\big(\log_e 8 - \log_e 2\big) \\ &= -\tfrac{1}{3}\log_e 4\end{aligned}$$

In **b**, $1-3\times 3=-8$ and $1-3\times 1=-2$, and $\log_e 8-\log_e 2=\log_e \frac{8}{2}=\log_e 4$.

c The denominator $x-3$ is zero at $x=3$, which lies between 1 and 4. This definite integral is meaningless because it crosses the asymptote at $x=3$.

d The interval $-4\le x\le -1$ is on the left-hand branch, so it does not cross $x=0$.

$$\begin{aligned}\int_{-4}^{-1} \frac{1}{x}\,dx &= \Big[\log_e|x|\Big]_{-4}^{-1} \\ &= \log_e|-1| - \log_e|-4| \\ &= 0 - \log_e 4 \\ &= -\log_e 4\end{aligned}$$

The integral is negative because the region is below the x-axis (the diagram in Concept 2).

Your turn.

a

$$\begin{aligned}\int_e^{e^4} \frac{2}{x}\,dx &= 2\Big[\log_e|x|\Big]_e^{e^4} \\ &= 2\big(\log_e e^4 - \log_e e\big) \\ &= 2(4-1) \\ &= 6\end{aligned}$$

b $(a = -2 \text{ and } b = 1)$

$$\begin{aligned}\int_1^2 \frac{1}{1-2x}\,dx &= -\tfrac{1}{2}\Big[\log_e|1-2x|\Big]_1^2 \\ &= -\tfrac{1}{2}\big(\log_e|-3| - \log_e|-1|\big) \\ &= -\tfrac{1}{2}\big(\log_e 3 - 0\big) \\ &= -\tfrac{1}{2}\log_e 3\end{aligned}$$

c The denominator $x-1$ is zero at $x=1$, which lies between 0 and 3. This definite integral is meaningless because it crosses the asymptote at $x=1$.

d The interval $-6 \le x \le -2$ is on the left-hand branch.

$$\begin{aligned}\int_{-6}^{-2} \frac{1}{x}\,dx &= \Big[\log_e|x|\Big]_{-6}^{-2} \\ &= \log_e|-2| - \log_e|-6| \\ &= \log_e 2 - \log_e 6 \\ &= -\log_e 3\end{aligned}$$

The last line uses $\log_e 2 - \log_e 6 = \log_e \frac{2}{6} = \log_e \frac{1}{3} = -\log_e 3$.

C3 Tutor demo.

a Let $u = x^2 + 5$. Then $u' = 2x$, so the top is the derivative of the bottom.

$$\begin{aligned}\int_0^2 \frac{2x}{x^2+5}\,dx &= \Big[\log_e(x^2+5)\Big]_0^2 \\ &= \log_e 9 - \log_e 5\end{aligned}$$

b Let $u = 1 - x^2$. Then $u' = -2x$. Make the top the derivative of the bottom:

$$\begin{aligned}\int_3^5 \frac{x}{1-x^2}\,dx &= -\tfrac{1}{2}\int_3^5 \frac{-2x}{1-x^2}\,dx \\ &= -\tfrac{1}{2}\Big[\log_e|1-x^2|\Big]_3^5 \\ &= -\tfrac{1}{2}\big(\log_e|-24| - \log_e|-8|\big) \\ &= -\tfrac{1}{2}\big(\log_e 24 - \log_e 8\big) \\ &= -\tfrac{1}{2}\log_e 3\end{aligned}$$

Here $1-5^2 = -24$, $1-3^2 = -8$ and $\log_e 24 - \log_e 8 = \log_e \frac{24}{8} = \log_e 3$. The asymptotes $x = \pm 1$ are not between 3 and 5.

c The denominator $8 - x^3$ is zero at $x = 2$, which lies between 0 and 3. This definite integral is meaningless because it crosses the asymptote at $x = 2$.

Your turn.

a Let $u = x^2 + 3$. Then $u' = 2x$, so the top is the derivative of the bottom.

$$\begin{aligned}\int_0^1 \frac{2x}{x^2+3}\,dx &= \Big[\log_e(x^2+3)\Big]_0^1 \\ &= \log_e 4 - \log_e 3\end{aligned}$$

b Let $u = 1 - x^2$. Then $u' = -2x$.

$$\begin{aligned}\int_2^4 \frac{x}{1-x^2}\,dx &= -\tfrac{1}{2}\int_2^4 \frac{-2x}{1-x^2}\,dx \\ &= -\tfrac{1}{2}\Big[\log_e|1-x^2|\Big]_2^4 \\ &= -\tfrac{1}{2}\big(\log_e|-15| - \log_e|-3|\big) \\ &= -\tfrac{1}{2}\big(\log_e 15 - \log_e 3\big) \\ &= -\tfrac{1}{2}\log_e 5\end{aligned}$$

c The denominator $4 - x^2$ is zero at $x = 2$, which lies between 0 and 3. This definite integral is meaningless because it crosses the asymptote at $x = 2$.

C4

Tutor demo.

a Here $\dfrac{3}{2-x} = 3 \times \dfrac{1}{-x+2}$, with $a = -1$.

Taking the primitive,

$f(x) = -3\ln|2-x| + C$.

Because $f(0) = 0$, $0 = -3\ln 2 + C$,

so $C = 3\ln 2$.

Hence $f(x) = 3\ln 2 - 3\ln|2-x|$.

b $x = 0$ and $x = 1$ are both on the left of the asymptote $x = 2$.

$f(1) = 3\ln 2 - 3\ln 1 = 3\ln 2$

Your turn.

a Here $\dfrac{4}{5-x} = 4 \times \dfrac{1}{-x+5}$, with $a = -1$.

Taking the primitive,

$f(x) = -4\ln|5-x| + C$.

Because $f(0) = 0$, $0 = -4\ln 5 + C$,

so $C = 4\ln 5$.

Hence $f(x) = 4\ln 5 - 4\ln|5-x|$.

b $x = 0$ and $x = 4$ are both on the left of the asymptote $x = 5$.

$f(4) = 4\ln 5 - 4\ln 1 = 4\ln 5$

C5

Tutor demo.

a By the product rule with $u = x$ and $v = \log_e x$, so $u' = 1$ and $v' = \dfrac{1}{x}$,

$$\begin{aligned}\frac{d}{dx}(x\log_e x - x) &= \left(\log_e x + x \times \frac{1}{x}\right) - 1 \\ &= \log_e x + 1 - 1 \\ &= \log_e x.\end{aligned}$$

b Reversing part **a**, $\displaystyle\int \log_e x\,dx = x\log_e x - x + C$.

c

$$\begin{aligned}\int_1^{e^2} \log_e x\,dx &= \Big[x\log_e x - x\Big]_1^{e^2} \\ &= \left(e^2\log_e e^2 - e^2\right) - \left(1\log_e 1 - 1\right) \\ &= \left(2e^2 - e^2\right) - (0 - 1) \\ &= e^2 + 1\end{aligned}$$

Your turn.

a and **b** As in the tutor demo, $\dfrac{d}{dx}(x\log_e x - x) = \log_e x + 1 - 1 = \log_e x$, so $\displaystyle\int \log_e x\,dx = x\log_e x - x + C$.

c

$$\begin{aligned}\int_e^{e^2} \log_e x \, dx &= \Big[x \log_e x - x\Big]_e^{e^2} \\ &= \left(e^2 \log_e e^2 - e^2\right) - \left(e \log_e e - e\right) \\ &= \left(2e^2 - e^2\right) - (e - e) \\ &= e^2\end{aligned}$$

Textbook Homework

1 **a** $2\log_e|x| + C$ **b** $\frac{1}{3}\log_e|x| + C$ **c** $\frac{4}{5}\log_e|x| + C$ **d** $\frac{3}{2}\log_e|x| + C$

2 **a** $\frac{1}{4}\log_e|4x+1| + C$ **b** $\frac{1}{5}\log_e|5x-3| + C$ **c** $2\log_e|3x+2| + C$
d $3\log_e|5x+1| + C$ **e** $\log_e|4x+3| + C$ **f** $-\log_e|3-x| + C$
g $-\frac{1}{2}\log_e|7-2x| + C$ **h** $\frac{4}{5}\log_e|5x-1| + C$ **i** $-4\log_e|1-3x| + C$

3 **a** $\log_e 5$ **b** $\log_e 3$ **c** $\log_e|-2| - \log_e|-8| = -2\log_e 2$
d The integral is meaningless because it runs across an asymptote at $x = 0$.
e $\frac{1}{2}(\log_e 8 - \log_e 2) = \log_e 2$ **f** $\frac{1}{5}(\log_e|-25| - \log_e|-75|) = -\frac{1}{5}\log_e 3$

4 **a** $\log_e 2 \doteq 0.6931$ **b** $\log_e 3 - \log_e 5 \doteq -0.5108$ **c** $-\frac{1}{2}\log_e 7 \doteq -0.9730$
d $\frac{3}{2}\log_e 3 \doteq 1.648$ **e** $\log_e \frac{5}{2} \doteq 0.9163$
f The integral is meaningless because it runs across an asymptote at $x = 5\frac{1}{2}$.

5 **a** 1 **b** 2 **c** 3 **d** $\frac{1}{2}$

6 **a** $x + \log_e|x| + C$ **b** $\frac{1}{5}x + \frac{3}{5}\log_e|x| + C$ **c** $\frac{1}{9}\log_e|x| - \frac{8}{9}x + C$
d $3x - 2\log_e|x| + C$ **e** $x^2 + x - 4\log_e|x| + C$
f $\frac{1}{3}x^3 - \log_e|x| - \frac{2}{x} + C$

7 **a** $\log_e|x^2-9| + C$ **b** $\log_e|3x^2+x| + C$ **c** $\log_e|x^2+x-3| + C$
d $\log_e|2+5x-3x^2| + C$ **g** $\log_e(1+e^x) + C$ **i** $\log_e(e^x + e^{-x}) + C$
The denominators in parts **g** and **i** are never negative, so the absolute value sign is unnecessary.

8 **a** $\frac{1}{3}\log_e|3x-k| + C$ **b** $\frac{1}{m}\log_e|mx-2| + C$ **c** $\log_e|px+q| + C$
d $\frac{A}{s}\log_e|sx-t| + C$

9 **a** $f(x) = x + 2\ln|x|$, $f(2) = 2 + 2\ln 2$
b $f(x) = x^2 + \frac{1}{3}\ln|x| + 1$, $f(2) = 5 + \frac{1}{3}\ln 2$
c $f(x) = 3x + \frac{5}{2}\ln|2x-1| - 3$, $f(2) = 3 + \frac{5}{2}\ln 3$
d $f(x) = 2x^3 + 5\ln|3x+2| - 2$, $f(2) = 14 + 5\ln 8$

10 **a** $f(x) = x + \ln|x| + \frac{1}{2}x^2$ **b** $g(x) = x^2 - 3\ln|x| + \frac{4}{x} - 6$

11 **a** $y = \frac{1}{4}(\log_e |x| + 2)$, $x = e^{-2}$ **b** $y = 2\log_e |x+1| + 1$
c $y = \log_e \left| \dfrac{x^2+5x+4}{10} \right| + 1$, $y(0) = \log_e \frac{4}{10} + 1$
d $y = 2\log_e |x| + x + C$, $y = 2\log_e |x| + x$, $y(2) = \log_e 4 + 2$
e $f(x) = 2 + x - \log_e |x|$, $f(e) = e + 1$

12 **a** $\log_e |x^3 - 5| + C$ **b** $\log_e |x^4 + x - 5| + C$ **c** $\frac{1}{4}\log_e |x^4 - 6x^2| + C$
d $\frac{1}{2}\log_e |5x^4 - 7x^2 + 8| + C$ **e** $2\log_e 2$ **f** $\log_e \dfrac{4(e+1)}{e+2}$

13 **a** (i) $y' = \log_e x$ (ii) $x\log_e x - x + C$ and $\dfrac{\sqrt{e}}{2}$
b (i) $y' = 4x\log_e x$ (ii) $\frac{1}{2}x^2\log_e x - \frac{1}{4}x^2 + C$ and $2\log_e 2 - 1 - \dfrac{e^2}{4}$
c $\dfrac{2\log_e x}{x}$ and $\dfrac{3}{8}$
d $\dfrac{d}{dx}\ln(\ln x) = \dfrac{1}{x\ln x}$, so the primitives are $\ln(\ln x) + C$ for $x > 1$ (in general, $\ln|\ln x| + C$)

14 **a** $a = e^5$ **b** $a = e^{-4}$ **c** $a = -e^2$ **d** $a = -e^{-1}$

15 **a** $\log_e(e^x + 1) + C$ **b** $\frac{1}{3}(e^3 - e^{-3}) + 2$ **c** $\dfrac{d}{dx}\, xe^x = (x+1)e^x$

16 The key to all this is that $\log_e |5x| = \log_e 5 + \log_e |x|$, so that $\log_e |x|$ and $\log_e |5x|$ differ only by a constant $\log_e 5$. Thus $C_2 = C_1 - \frac{1}{5}\log_e 5$, and because C_1 and C_2 are arbitrary constants, it does not matter at all. In particular, in a definite integral, adding a constant doesn't change the answer, because it cancels out when we take $F(b) - F(a)$.

17 $$y = \begin{cases} \log_e x + 1, & \text{for } x > 0, \\ \log_e(-x) + 2, & \text{for } x < 0. \end{cases}$$

18 **d** (i) $\log_e \frac{3}{2} \doteq 0.41$ (ii) $\log_e 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$
e $\log_e(1-x) = -x - \dfrac{x^2}{2} - \dfrac{x^3}{3} - \dfrac{x^4}{4} - \cdots$, $\log_e \frac{1}{2} \doteq -0.69$
f Using $x = \frac{1}{2}$, $\log_e 3 \doteq 1.0986$.

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 6I | Integration of the reciprocal function

Past-Paper Homework - Worked Solutions

PP 1 **North Sydney Boys High School, 2022, HSC Course Assessment Task 3-Trial Examination, Mathematics Extension 1, Q13(a)(i)**

The integrand needs $\ln x$, so $x > 0$ (and $x \neq 1$, so that $\ln x \neq 0$). Split the fraction into two separate fractions:

$$\begin{aligned}\frac{1-\ln x}{x\ln x} &= \frac{1}{x\ln x} - \frac{\ln x}{x\ln x} \\ &= \frac{1}{x\ln x} - \frac{1}{x}.\end{aligned}$$

For the first fraction, let $u = \ln x$. Then $u' = \dfrac{1}{x}$, so

$$\frac{1}{x\ln x} = \frac{\frac{1}{x}}{\ln x} = \frac{u'}{u}, \quad \text{and} \quad \int \frac{u'}{u}\,dx = \ln|u| = \ln|\ln x|.$$

Hence

$$\begin{aligned}\int \frac{1-\ln x}{x\ln x}\,dx &= \int \left(\frac{1}{x\ln x} - \frac{1}{x}\right) dx \\ &= \ln|\ln x| - \ln x + C,\end{aligned}$$

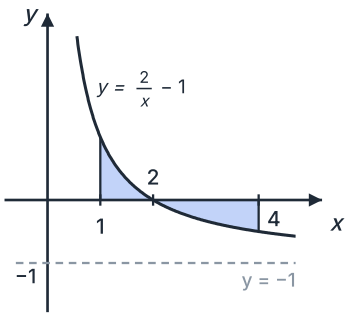
for some constant C. Here $\ln|x| = \ln x$ because $x > 0$. Using a log law, this can also be written as $\ln\left|\dfrac{\ln x}{x}\right| + C$.

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 6J | Applications of integration of 1/x

Lesson Examples

C1 **Tutor demo.**



a The graph of $y = \dfrac{2}{x}$ has asymptotes $x = 0$ and $y = 0$. Move it down 1 unit.
The asymptotes are $x = 0$ and $y = -1$.
When $y = 0$, $\dfrac{2}{x} = 1$, so $x = 2$.

b The region crosses the x-axis at $x = 2$, so split the integral there. Here $x > 0$, so $|x| = x$.

$$\begin{aligned}\int_1^2 \left(\frac{2}{x} - 1\right) dx &= \Big[2\log_e x - x\Big]_1^2 \\ &= \left(2\log_e 2 - 2\right) - \left(2\log_e 1 - 1\right) \\ &= \left(2\log_e 2 - 2\right) - \left(0 - 1\right) \\ &= 2\log_e 2 - 1.\end{aligned}$$

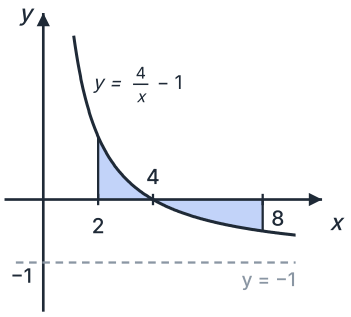
This is positive (about 0.386), because this part is above the x-axis.

$$\begin{aligned}\int_2^4 \left(\frac{2}{x} - 1\right) dx &= \Big[2\log_e x - x\Big]_2^4 \\ &= \left(2\log_e 4 - 4\right) - \left(2\log_e 2 - 2\right) \\ &= 2\log_e 4 - 2\log_e 2 - 2 \\ &= 2\log_e 2 - 2.\end{aligned}$$

This is negative (about -0.614), because this part is below the x-axis. In the last line, $2\log_e 4 - 2\log_e 2 = 2\log_e \frac{4}{2} = 2\log_e 2$.

$$\begin{aligned}\textbf{Hence area} &= \left(2\log_e 2 - 1\right) + \left(2 - 2\log_e 2\right) \\ &= 1 \textbf{ square unit.}\end{aligned}$$

Your turn.



a The graph of $y = \dfrac{4}{x}$ has asymptotes $x = 0$ and $y = 0$. Move it down 1 unit.
The asymptotes are $x = 0$ and $y = -1$.
When $y = 0$, $\dfrac{4}{x} = 1$, so $x = 4$.

b The region crosses the x-axis at $x = 4$, so split the integral there. Here $x > 0$, so $|x| = x$.

$$\begin{aligned}\int_2^4 \left(\frac{4}{x} - 1\right) dx &= \Big[4\log_e x - x\Big]_2^4 \\ &= \left(4\log_e 4 - 4\right) - \left(4\log_e 2 - 2\right) \\ &= 4\log_e 4 - 4\log_e 2 - 2 \\ &= 4\log_e 2 - 2.\end{aligned}$$

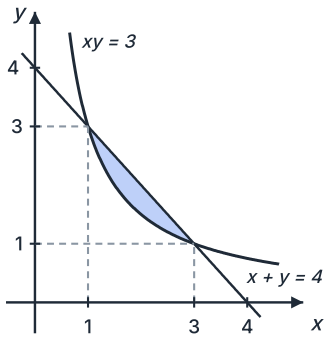
This is positive (about 0.773), because this part is above the x-axis.

$$\begin{aligned}\int_4^8 \left(\frac{4}{x} - 1\right) dx &= \Big[4\log_e x - x\Big]_4^8 \\ &= \left(4\log_e 8 - 8\right) - \left(4\log_e 4 - 4\right) \\ &= 4\log_e 8 - 4\log_e 4 - 4 \\ &= 4\log_e 2 - 4.\end{aligned}$$

This is negative (about -1.227), because this part is below the x-axis. Here $4\log_e 4 - 4\log_e 2 = 4\log_e \frac{4}{2} = 4\log_e 2$ and $4\log_e 8 - 4\log_e 4 = 4\log_e \frac{8}{4} = 4\log_e 2$.

$$\begin{aligned}\text{Hence area} &= \left(4\log_e 2 - 2\right) + \left(4 - 4\log_e 2\right) \\ &= 2 \text{ square units.}\end{aligned}$$

C2 Tutor demo.



a Substituting the line $y = 4 - x$ into the hyperbola,

$x(4 - x) = 3$, so $4x - x^2 = 3$,

$x^2 - 4x + 3 = 0$

$(x - 1)(x - 3) = 0$

$x = 1$ or 3,

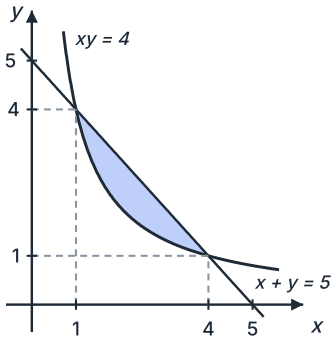
so the curves meet at $(1, 3)$ and $(3, 1)$.

b The hyperbola $xy = 3$ has both axes as asymptotes. The line $x + y = 4$ has x-intercept $(4, 0)$ and y-intercept $(0, 4)$.

c The line is on top: at $x = 2$, the line gives 2 and the hyperbola gives $1\frac{1}{2}$.

$$\begin{aligned}\text{Area} &= \int_1^3 \left((4-x) - \frac{3}{x}\right) dx \\ &= \left[4x - \tfrac{1}{2}x^2 - 3\log_e|x|\right]_1^3 \\ &= \left(12 - 4\tfrac{1}{2} - 3\log_e 3\right) - \left(4 - \tfrac{1}{2} - 3\log_e 1\right) \\ &= \left(7\tfrac{1}{2} - 3\log_e 3\right) - \left(3\tfrac{1}{2} - 0\right) \\ &= \left(4 - 3\log_e 3\right) \text{ square units} \\ &\doteq 0.704 \text{ square units.}\end{aligned}$$

Your turn.



a Substituting the line $y = 5 - x$ into the hyperbola,
$x(5-x) = 4$, so $5x - x^2 = 4$,
$x^2 - 5x + 4 = 0$
$(x-1)(x-4) = 0$
$x = 1$ or 4,
so the curves meet at $(1, 4)$ and $(4, 1)$.

b The hyperbola $xy = 4$ has both axes as asymptotes. The line $x + y = 5$ has x-intercept $(5, 0)$ and y-intercept $(0, 5)$.

c The line is on top: at $x = 2$, the line gives 3 and the hyperbola gives 2.

$$\begin{aligned}\text{Area} &= \int_1^4 \left((5-x) - \frac{4}{x}\right) dx \\ &= \left[5x - \tfrac{1}{2}x^2 - 4\log_e|x|\right]_1^4 \\ &= \left(20 - 8 - 4\log_e 4\right) - \left(5 - \tfrac{1}{2} - 4\log_e 1\right) \\ &= \left(12 - 4\log_e 4\right) - \left(4\tfrac{1}{2} - 0\right) \\ &= \left(7\tfrac{1}{2} - 4\log_e 4\right) \text{ square units} \\ &\doteq 1.955 \text{ square units.}\end{aligned}$$

Textbook Homework

1 **b** $e \doteq 2.7$

2 **i** $\log_e 5 \doteq 1.609$ u^2 **ii** 1 u^2 **iii** $2\log_e 2 \doteq 1.386$ u^2

3 **a** $(\log_e 3 - \log_e 2)$ square units **b** $\log_e 2 - \log_e \frac{1}{2} = 2\log_e 2$ square units

4 **a** $\frac{1}{3}(\log_e 5 - \log_e 2)$ u^2 **b** 9 u^2 **c** $\left(2\log_e 2 + \frac{15}{8}\right)$ u^2 **d** $\left(\log_e 3 + 8\frac{2}{3}\right)$ u^2

5 **a** $(6 - 3\log_e 3)$ u^2 **b** $(4 - \log_e 3)$ u^2

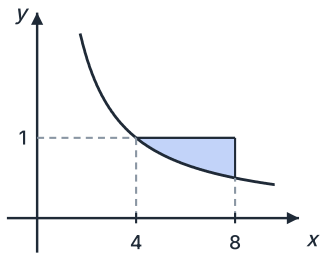
6 **a** $\left(3\frac{3}{4} - 2\log_e 4\right)$ u^2 **b** $\left(3\frac{3}{4} - 2\log_e 4\right)$ u^2

7 **a** $2\log_e 2$ u^2 **b** $(1 - \log_e 2)$ u^2

8 **a** $\log_e 4$ u^2 **b** $(6 - 3\log_e 3)$ u^2

9 **a** $\frac{1}{2}$ u^2 **b** $2\log_e \frac{4}{3}$ u^2

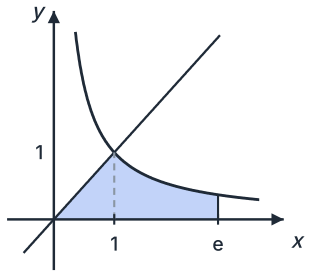
10 **a** **b** $(4 - 4\log_e 2)$ u^2



11 **a** $\left(\frac{1}{3}, 3\right)$ and $(1, 1)$ **b** $\left(\frac{4}{3} - \log_e 3\right)$ u^2

12 **a** $2x$, $\frac{1}{2}\log_e 5 \doteq 0.805$ u^2 **b** $2(x+1)$, $\frac{1}{2}\log_e 2$ u^2

13 **a** **b** $\frac{3}{2}$ u^2



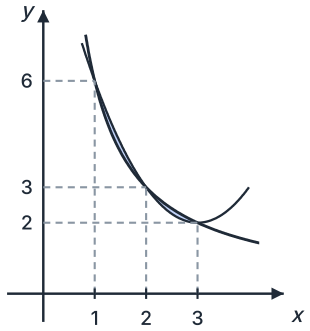
14 **a** 3.9828 square units

b $\dfrac{d}{dx}\left(x\log_e x\right) = \log_e x + 1$, area $= 5\log_e 5 - 4 \doteq 4.0472$ square units

c The estimate is less. The curve is concave down, so the chords are below the curve.

15 **b** $(2 - \log_e 3)$ u^2

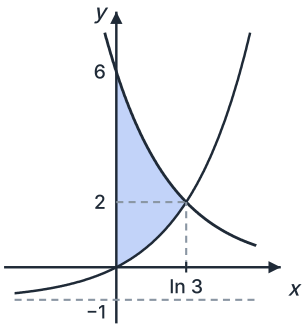
16 **b** **c** $\left(2 - 6\log_e \frac{4}{3}\right)$ u^2



17 **a** The upper rectangle has height 2^{-n}, the lower rectangle has height 2^{-n-1}, and both rectangles have width $2^{n+1} - 2^n = 2^n$.

18 **b** $(\log_e 3,\ 2)$ **d** $(2 + \log_e 3)$ u^2

c



19 **a** $e - 2$ square units **b** e^{-1} square units **c** $e - 2 + e^{-1}$ square units

20 $2\log_e\left(\sqrt{x} + 1\right) + C$

21 **b i** $\log_e\left(1 + \sqrt{2}\right)$ **ii** $\log_e\left(2 + \sqrt{3}\right)$

23 **a i** The region is above the t-axis, and contained within the rectangle $ABCO$.
ii The rectangle $ABCO$ has width $\sqrt{x}$ and height 1, so its area is $\sqrt{x}$.

$$0 < \Big[\log_e t\Big]_1^{\sqrt{x}} < \sqrt{x}$$
$$0 < \log_e \sqrt{x} - \log_e 1 < \sqrt{x}$$
$$0 < \tfrac{1}{2}\log_e x < \sqrt{x} \qquad \left(\times \tfrac{2}{x}\right)$$
$$0 < \frac{\log_e x}{x} < \frac{2}{\sqrt{x}}.$$

iii The third term has limit 0 as $x \to \infty$, so the result follows by the sandwiching principle.

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 6K | Calculus with other bases

Lesson Examples

C1 **Tutor demo.**

a $\log_3 x = \dfrac{\log_e x}{\log_e 3}$

b i $\log_3 40 = \dfrac{\log_e 40}{\log_e 3} = \dfrac{3.688\,879\ldots}{1.098\,612\ldots} \doteq 3.3578$

ii $\log_3 2 = \dfrac{\log_e 2}{\log_e 3} = \dfrac{0.693\,147\ldots}{1.098\,612\ldots} \doteq 0.6309$

iii $\log_3 0.05 = \dfrac{\log_e 0.05}{\log_e 3} = \dfrac{-2.995\,732\ldots}{1.098\,612\ldots} \doteq -2.7268$

c i $3^{3.3578} \doteq 40$ ii $3^{0.6309} \doteq 2$ iii $3^{-2.7268} \doteq 0.05$

Your turn.

a $\log_7 x = \dfrac{\log_e x}{\log_e 7}$

b i $\log_7 50 = \dfrac{\log_e 50}{\log_e 7} = \dfrac{3.912\,023\ldots}{1.945\,910\ldots} \doteq 2.0104$

ii $\log_7 3 = \dfrac{\log_e 3}{\log_e 7} = \dfrac{1.098\,612\ldots}{1.945\,910\ldots} \doteq 0.5646$

iii $\log_7 0.2 = \dfrac{\log_e 0.2}{\log_e 7} = \dfrac{-1.609\,437\ldots}{1.945\,910\ldots} \doteq -0.8271$

c i $7^{2.0104} \doteq 50$ ii $7^{0.5646} \doteq 3$ iii $7^{-0.8271} \doteq 0.2$

C2 **Tutor demo.**

1a Here $y = \log_3 x = \dfrac{\log_e x}{\log_e 3}$.

Because $\log_e 3$ is a constant,

$$\frac{dy}{dx} = \frac{1}{x} \times \frac{1}{\log_e 3} = \frac{1}{x \log_e 3}.$$

1b Here $y = 4\log_a x = \dfrac{4\log_e x}{\log_e a}$.

Because $\dfrac{4}{\log_e a}$ is a constant,

$$\frac{dy}{dx} = \frac{4}{\log_e a} \times \frac{1}{x} = \frac{4}{x \log_e a}.$$

2a $\dfrac{d}{dx}\log_5 x = \dfrac{1}{x \log_e 5}$ **2b** $\dfrac{d}{dx}\log_{1.08} x = \dfrac{1}{x \log_e 1.08}$

Your turn.

1a Here $y = \log_7 x = \dfrac{\log_e x}{\log_e 7}$.

Because $\log_e 7$ is a constant,

$$\frac{dy}{dx} = \frac{1}{x} \times \frac{1}{\log_e 7}$$
$$= \frac{1}{x \log_e 7}.$$

1b Here $y = 2\log_a x = \dfrac{2\log_e x}{\log_e a}$.

Because $\dfrac{2}{\log_e a}$ is a constant,

$$\frac{dy}{dx} = \frac{2}{\log_e a} \times \frac{1}{x}$$
$$= \frac{2}{x \log_e a}.$$

2a $\dfrac{d}{dx}\log_4 x = \dfrac{1}{x \log_e 4}$ **2b** $\dfrac{d}{dx}\log_{1.1} x = \dfrac{1}{x \log_e 1.1}$

C3 Tutor demo.

a When $y = 0$, $\log_3 x = 0$, so $x = 3^0 = 1$. The x-intercept is $(1, 0)$.

Differentiating, $y' = \dfrac{1}{x \log_e 3}$,

so when $x = 1$, $y' = \dfrac{1}{1 \times \log_e 3} = \dfrac{1}{\log_e 3}$, as required.

b The tangent passes through $(1, 0)$ with gradient $\dfrac{1}{\log_e 3}$, so

$$y - 0 = \frac{1}{\log_e 3}(x - 1)$$
$$y = \frac{x - 1}{\log_e 3}.$$

c By part **a** with base a, the gradient at the x-intercept is $\dfrac{1}{\log_e a}$. So

$$\frac{1}{\log_e a} = 2$$
$$\log_e a = \tfrac{1}{2}$$
$$a = e^{\frac{1}{2}} = \sqrt{e}.$$

Your turn.

a When $y = 0$, $\log_5 x = 0$, so $x = 5^0 = 1$. The x-intercept is $(1, 0)$.

Differentiating, $y' = \dfrac{1}{x \log_e 5}$,

so when $x = 1$, $y' = \dfrac{1}{1 \times \log_e 5} = \dfrac{1}{\log_e 5}$, as required.

b The tangent passes through $(1, 0)$ with gradient $\dfrac{1}{\log_e 5}$, so

$$\begin{aligned} y - 0 &= \frac{1}{\log_e 5}(x - 1) \\ y &= \frac{x - 1}{\log_e 5}. \end{aligned}$$

c The gradient at the x-intercept is $\dfrac{1}{\log_e a}$. So

$$\begin{aligned} \frac{1}{\log_e a} &= \tfrac{1}{2} \\ \log_e a &= 2 \\ a &= e^2. \end{aligned}$$

C4 Tutor demo.

a $3 = e^{\log_e 3}$

b
$$\begin{aligned} 3^x &= \left(e^{\log_e 3}\right)^x \\ &= e^{x \log_e 3} \end{aligned}$$

c
$$\begin{aligned} 7^{-x} &= \left(e^{\log_e 7}\right)^{-x} \\ &= e^{-x \log_e 7} \end{aligned}$$

d
$$\begin{aligned} 10^{2x} &= \left(e^{\log_e 10}\right)^{2x} \\ &= e^{2x \log_e 10} \end{aligned}$$

Your turn.

a $5 = e^{\log_e 5}$

b
$$\begin{aligned} 5^x &= \left(e^{\log_e 5}\right)^x \\ &= e^{x \log_e 5} \end{aligned}$$

c
$$\begin{aligned} 2^{-x} &= \left(e^{\log_e 2}\right)^{-x} \\ &= e^{-x \log_e 2} \end{aligned}$$

d
$$\begin{aligned} 6^{3x} &= \left(e^{\log_e 6}\right)^{3x} \\ &= e^{3x \log_e 6} \end{aligned}$$

C5 Tutor demo.

a

Here $y = 3^x$.

Using the standard form, $y' = 3^x \log_e 3$.

Hence when $x = 0$,
$$\begin{aligned} y' &= 3^0 \times \log_e 3 \\ &= \log_e 3 \\ &\doteq 1.10. \end{aligned}$$

b $3^x = \left(e^{\log_e 3}\right)^x = e^{x \log_e 3}$, so with $k = \log_e 3$,

$$\begin{aligned} \int 3^x \, dx &= \int e^{x \log_e 3} \, dx \\ &= \frac{e^{x \log_e 3}}{\log_e 3} + C \\ &= \frac{3^x}{\log_e 3} + C. \end{aligned}$$

Your turn.

a

Here $y = 5^x$.

Using the standard form, $y' = 5^x \log_e 5$.

Hence when $x = 0$, $y' = 5^0 \times \log_e 5$

$= \log_e 5$

$\doteq 1.61$.

b $5^x = \left(e^{\log_e 5}\right)^x = e^{x \log_e 5}$, so with $k = \log_e 5$,

$$\int 5^x \, dx = \int e^{x \log_e 5} \, dx$$

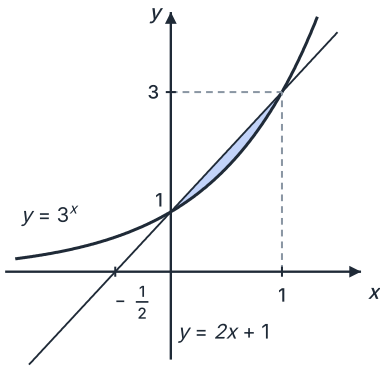
$$= \frac{e^{x \log_e 5}}{\log_e 5} + C$$

$$= \frac{5^x}{\log_e 5} + C.$$

C6 Tutor demo.

a When $x = 0$: $2x + 1 = 1$ and $3^0 = 1$. When $x = 1$: $2x + 1 = 3$ and $3^1 = 3$. So both graphs pass through $A(0, 1)$ and $B(1, 3)$.

c At $x = \frac{1}{2}$, $2x + 1 = 2$ and $3^{\frac{1}{2}} = \sqrt{3} \doteq 1.732$, **b** so the line is the upper curve.



$$\text{Area} = \int_0^1 \left(2x + 1 - 3^x\right) dx$$

$$= \left[x^2 + x - \frac{3^x}{\log_e 3}\right]_0^1$$

$$= \left(1 + 1 - \frac{3}{\log_e 3}\right) - \left(0 + 0 - \frac{1}{\log_e 3}\right)$$

$$= 2 - \frac{3}{\log_e 3} + \frac{1}{\log_e 3}$$

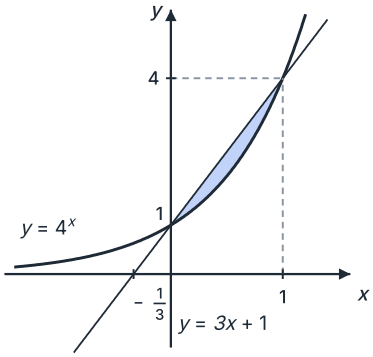
$$= 2 - \frac{2}{\log_e 3} \text{ square units}$$

$$\doteq 0.1795 \text{ square units.}$$

Your turn.

a When $x = 0$: $3x + 1 = 1$ and $4^0 = 1$. When $x = 1$: $3x + 1 = 4$ and $4^1 = 4$. So both graphs pass through $A(0, 1)$ and $B(1, 4)$.

c At $x = \frac{1}{2}$, $3x + 1 = 2.5$ and $4^{\frac{1}{2}} = 2$, so the line is the upper curve. **b**



$$\begin{aligned}
\text{Area} &= \int_0^1 \left(3x + 1 - 4^x\right) dx \\
&= \left[\tfrac{3}{2}x^2 + x - \frac{4^x}{\log_e 4}\right]_0^1 \\
&= \left(\tfrac{3}{2} + 1 - \frac{4}{\log_e 4}\right) - \left(0 + 0 - \frac{1}{\log_e 4}\right) \\
&= 2\tfrac{1}{2} - \frac{4}{\log_e 4} + \frac{1}{\log_e 4} \\
&= 2\tfrac{1}{2} - \frac{3}{\log_e 4} \text{ square units} \\
&\doteq 0.3360 \text{ square units.}
\end{aligned}$$

Textbook Homework

1 **a** $\dfrac{\log_e 3}{\log_e 2} \doteq 1.58$ **b** $\dfrac{\log_e 10}{\log_e 2} \doteq 3.32$ **c** $\dfrac{\log_e 26}{\log_e 5} \doteq 2.02$
d $\dfrac{\log_e 0.0047}{\log_e 3} \doteq -4.88$

2 **a** $y = \dfrac{\log_e x}{\log_e 2}$, $y' = \dfrac{1}{x \log_e 2}$ **b** $y = \dfrac{\log_e x}{\log_e 10}$, $y' = \dfrac{1}{x \log_e 10}$
c $y = \dfrac{3 \log_e x}{\log_e 5}$, $y' = \dfrac{3}{x \log_e 5}$

3 **a** $y' = \dfrac{1}{x \log_e 3}$ **b** $y' = \dfrac{1}{x \log_e 7}$ **c** $y' = \dfrac{5}{x \log_e 6}$

4 **a** $y = e^{x \log_e 3}$, $y' = 3^x \log_e 3$ **b** $y = e^{x \log_e 4}$, $y' = 4^x \log_e 4$
c $y = e^{x \log_e 2}$, $y' = 2^x \log_e 2$

5 **a** $y' = 10^x \log_e 10$ **b** $y' = 8^x \log_e 8$ **c** $y' = 3 \times 5^x \log_e 5$

6 **a** $\dfrac{2^x}{\log_e 2} + C$ **b** $\dfrac{6^x}{\log_e 6} + C$ **c** $\dfrac{7^x}{\log_e 7} + C$ **d** $\dfrac{3^x}{\log_e 3} + C$

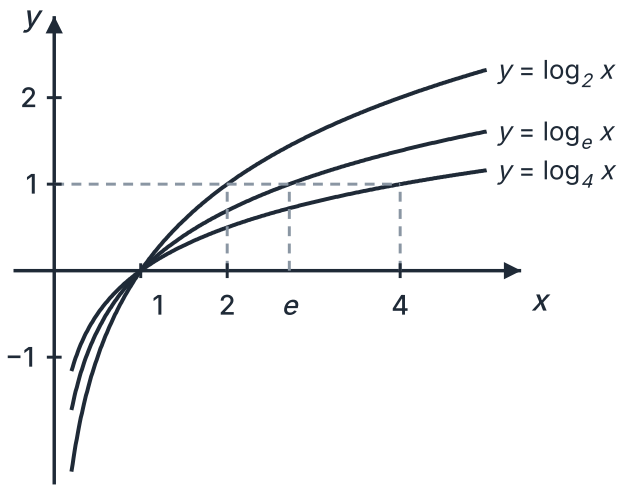
7 a $\dfrac{1}{\log_e 2} \doteq 1.443$ b $\dfrac{2}{\log_e 3} \doteq 1.820$ c $\dfrac{24}{5\log_e 5} \doteq 2.982$

d $\dfrac{15}{\log_e 4} \doteq 10.82$

8 a

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
$\log_2 x$	-2	-1	0	1	2
$\log_e x$	-1.39	-0.69	0	0.69	1.39
$\log_4 x$	-1	-0.5	0	0.5	1

b



9 a $y' = \dfrac{1}{x\log_e 2}$, gradient $\dfrac{1}{\log_e 2}$ b $y = \dfrac{1}{\log_e 2}(x-1)$

c i $y = \dfrac{1}{\log_e 3}(x-1)$ ii $y = \dfrac{1}{\log_e 5}(x-1)$

10 a $\dfrac{6}{\log_e 2} \doteq 8.6562$ b $2 + \dfrac{8}{3\log_e 3} \doteq 4.4273$ c $\dfrac{99}{\log_e 10} - 20 \doteq 22.9952$

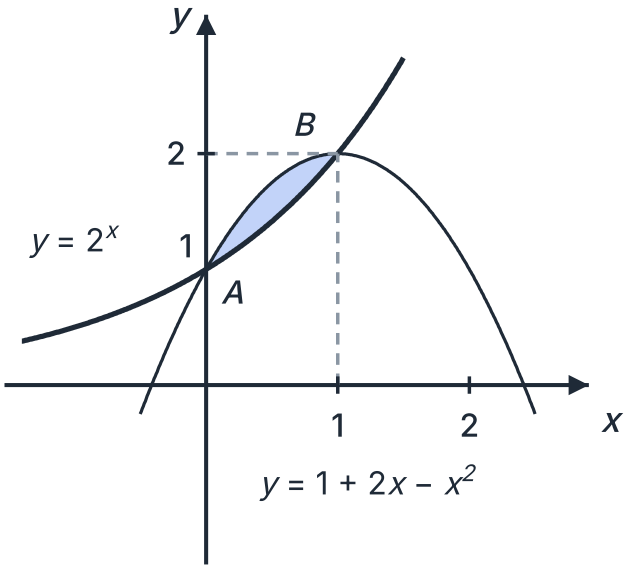
11 $y = \dfrac{\log_e x}{\log_e 10}$, $y' = \dfrac{1}{x\log_e 10}$

a $\dfrac{1}{10\log_e 10}$ b $x - 10y\log_e 10 + 10(\log_e 10 - 1) = 0$ c $x = \dfrac{1}{\log_e 10}$

12 **a** $y = \dfrac{1}{\log_e 2}\left(\dfrac{x}{3} - 1 + \log_e 3\right)$, $y = \dfrac{x}{3} - 1 + \log_e 3$, $y = \dfrac{1}{\log_e 4}\left(\dfrac{x}{3} - 1 + \log_e 3\right)$

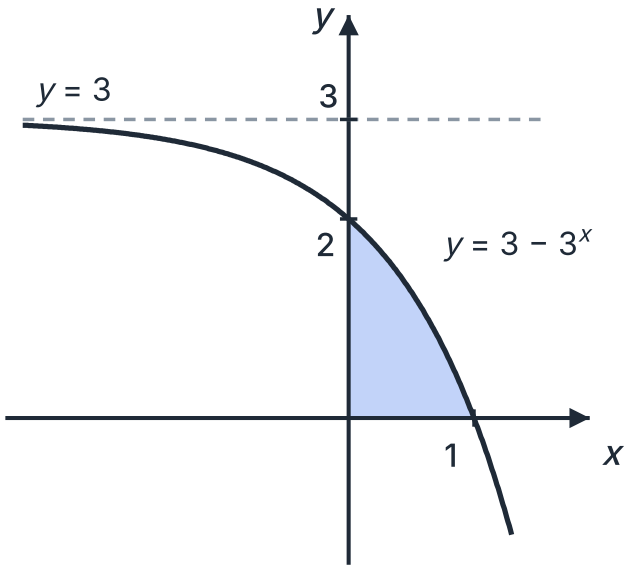
b They all meet the x-axis at $(3 - 3\log_e 3,\ 0)$.

13 **b** $\left(\dfrac{5}{3} - \dfrac{1}{\log_e 2}\right)$ square units



14 The intercepts are $(0, 7)$ and $(3, 0)$. The area is $\left(24 - \dfrac{7}{\log_e 2}\right)$ square units.

15 **a**



b $\left(3 - \dfrac{2}{\log_e 3}\right)$ square units

16 **b** $\displaystyle\int_{-\frac{1}{2}}^{0} \left(x + 1 - 4^x\right) dx$ **c** $\dfrac{3}{8} - \dfrac{1}{2\log_e 4}$

18

a $\dfrac{d}{dx}\left(x\log_e x - x\right) = \log_e x$, so $\displaystyle\int \log_e x\,dx = x\log_e x - x + C$

b $10 - \dfrac{9}{\log_e 10}$

19

a i $y' = \dfrac{1}{x\log_e 3}$ **ii** $y' = \dfrac{2}{(2x+3)\log_e 7}$ **iii** $y' = -\dfrac{45}{(4-9x)\log_e 6}$

b i $y' = 10^x \log_e 10$ **ii** $y' = 4 \times 8^{4x-3}\log_e 8$ **iii** $y' = -21 \times 5^{2-7x}\log_e 5$

c i $\dfrac{3^{5x}}{5\log_e 3} + C$ **ii** $\dfrac{6^{2x+7}}{2\log_e 6} + C$ **iii** $-\dfrac{5 \times 7^{4-9x}}{9\log_e 7} + C$

20

a $y = e^{kx}$ and $y = \dfrac{1}{k}\log_e x$

b The functions $y = a^x$ and $y = \log_a x$ are inverse, so they are symmetric in the line $y = x$. The common tangent is therefore the line $y = x$, which has gradient 1. (This argument would be invalid if there were more than one intersection point.)

c $ke^{kx} = 1$ and $\dfrac{1}{kx} = 1$

d $k = \dfrac{1}{e}$, $a = e^{\frac{1}{e}}$

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 7D | Integrating the trigonometric functions

Lesson Examples

C1 **Tutor demo.** The region is entirely above the x-axis, so

$$\begin{aligned}\text{area} &= \int_0^{\frac{\pi}{2}} \cos x \, dx \\ &= \Big[\sin x\Big]_0^{\frac{\pi}{2}} \\ &= \sin \tfrac{\pi}{2} - \sin 0 \\ &= 1 - 0 \\ &= 1\end{aligned}$$

The area is 1 square unit.

Your turn. The region is entirely above the x-axis, so

$$\begin{aligned}\text{area} &= \int_0^{\frac{\pi}{2}} \sin x \, dx \\ &= \Big[-\cos x\Big]_0^{\frac{\pi}{2}} \\ &= -\cos \tfrac{\pi}{2} + \cos 0 \\ &= 0 + 1 \\ &= 1\end{aligned}$$

The area is 1 square unit.

C2 **Tutor demo.**

a

$$\begin{aligned}\int_0^{2\pi} \cos x \, dx &= \Big[\sin x\Big]_0^{2\pi} \\ &= \sin 2\pi - \sin 0 \\ &= 0 - 0 \\ &= 0\end{aligned}$$

b

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \sec^2 x \, dx &= \Big[\tan x\Big]_0^{\frac{\pi}{4}} \\ &= \tan \tfrac{\pi}{4} - \tan 0 \\ &= 1 - 0 \\ &= 1\end{aligned}$$

c This integral is meaningless because it crosses the asymptote at $x = \frac{\pi}{2}$, which lies between $\frac{\pi}{3}$ and $\frac{2\pi}{3}$.

Your turn.

a

$$\begin{aligned}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos x \, dx &= \Big[\sin x\Big]_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \\ &= \sin \tfrac{3\pi}{2} - \sin \tfrac{\pi}{2} \\ &= -1 - 1 \\ &= -2\end{aligned}$$

b

$$\begin{aligned}\int_0^{\frac{\pi}{6}} \sec^2 x \, dx &= \Big[\tan x\Big]_0^{\frac{\pi}{6}} \\ &= \tan \tfrac{\pi}{6} - \tan 0 \\ &= \tfrac{1}{\sqrt{3}} - 0 \\ &= \tfrac{1}{\sqrt{3}}\end{aligned}$$

c This integral is meaningless because it crosses the asymptote at $x = \frac{\pi}{2}$, which lies between $\frac{\pi}{6}$ and $\frac{5\pi}{6}$.

C3 Tutor demo.

a $(a = 2)$

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \cos 2x \, dx &= \tfrac{1}{2}\Big[\sin 2x\Big]_0^{\frac{\pi}{4}} \\ &= \tfrac{1}{2}\big(\sin \tfrac{\pi}{2} - \sin 0\big) \\ &= \tfrac{1}{2}(1 - 0) \\ &= \tfrac{1}{2}\end{aligned}$$

b $(a = \frac{1}{3})$

$$\begin{aligned}\int_0^{\pi} \sin \tfrac{1}{3}x \, dx &= -3\Big[\cos \tfrac{1}{3}x\Big]_0^{\pi} \\ &= -3\big(\cos \tfrac{\pi}{3} - \cos 0\big) \\ &= -3\big(\tfrac{1}{2} - 1\big) \\ &= \tfrac{3}{2}\end{aligned}$$

c $(a = 2,\ b = \pi)$

$$\begin{aligned}\int_0^{\frac{\pi}{6}} \sec^2(2x + \pi) \, dx &= \tfrac{1}{2}\Big[\tan(2x + \pi)\Big]_0^{\frac{\pi}{6}} \\ &= \tfrac{1}{2}\big(\tan \tfrac{4\pi}{3} - \tan \pi\big) \\ &= \tfrac{1}{2}\big(\sqrt{3} - 0\big) \\ &= \tfrac{\sqrt{3}}{2}\end{aligned}$$

In **c**, $2 \times \frac{\pi}{6} + \pi = \frac{4\pi}{3}$, which is in quadrant 3 with related angle $\frac{\pi}{3}$, so $\tan \frac{4\pi}{3} = \tan \frac{\pi}{3} = \sqrt{3}$.

Your turn.

a $(a = 4)$

$$\begin{aligned}\int_0^{\frac{\pi}{8}} \cos 4x \, dx &= \tfrac{1}{4}\Big[\sin 4x\Big]_0^{\frac{\pi}{8}} \\ &= \tfrac{1}{4}\big(\sin \tfrac{\pi}{2} - \sin 0\big) \\ &= \tfrac{1}{4}(1 - 0) \\ &= \tfrac{1}{4}\end{aligned}$$

b $(a = \frac{1}{3})$

$$\begin{aligned}\int_0^{\frac{3\pi}{2}} \sin \tfrac{1}{3}x \, dx &= -3\Big[\cos \tfrac{1}{3}x\Big]_0^{\frac{3\pi}{2}} \\ &= -3\big(\cos \tfrac{\pi}{2} - \cos 0\big) \\ &= -3(0 - 1) \\ &= 3\end{aligned}$$

c $(a = 3,\ b = \pi)$

$$\begin{aligned}\int_0^{\frac{\pi}{12}} \sec^2(3x + \pi) \, dx &= \tfrac{1}{3}\Big[\tan(3x + \pi)\Big]_0^{\frac{\pi}{12}} \\ &= \tfrac{1}{3}\big(\tan \tfrac{5\pi}{4} - \tan \pi\big) \\ &= \tfrac{1}{3}(1 - 0) \\ &= \tfrac{1}{3}\end{aligned}$$

C4 Tutor demo.

a Let $u = \sin 2x$. Then $u' = 2\cos 2x$.

$$\begin{aligned}\int \cot 2x \, dx &= \int \frac{\cos 2x}{\sin 2x} \, dx \\ &= \tfrac{1}{2}\int \frac{2\cos 2x}{\sin 2x} \, dx \\ &= \tfrac{1}{2}\log_e |\sin 2x| + C\end{aligned}$$

b Let $u = \cos 2x$. Then $u' = -2\sin 2x$.

$$\begin{aligned}\int \tan 2x \, dx &= \int \frac{\sin 2x}{\cos 2x} \, dx \\ &= -\tfrac{1}{2}\int \frac{-2\sin 2x}{\cos 2x} \, dx \\ &= -\tfrac{1}{2}\log_e |\cos 2x| + C\end{aligned}$$

Your turn.

a Let $u = \sin 3x$. Then $u' = 3\cos 3x$.

$$\begin{aligned}\int \cot 3x \, dx &= \int \frac{\cos 3x}{\sin 3x} \, dx \\ &= \tfrac{1}{3}\int \frac{3\cos 3x}{\sin 3x} \, dx \\ &= \tfrac{1}{3}\log_e |\sin 3x| + C\end{aligned}$$

b Let $u = \cos 3x$. Then $u' = -3\sin 3x$.

$$\begin{aligned}\int \tan 3x \, dx &= \int \frac{\sin 3x}{\cos 3x} \, dx \\ &= -\tfrac{1}{3}\int \frac{-3\sin 3x}{\cos 3x} \, dx \\ &= -\tfrac{1}{3}\log_e |\cos 3x| + C\end{aligned}$$

C5 Tutor demo.

1 Taking the primitive,
$y = -\cos x + C$.
When $x = 0$, $y = 2$, so
$2 = -\cos 0 + C$,
so $2 = -1 + C$ and $C = 3$.
Hence $f(x) = 3 - \cos x$.
Then
$f\left(\frac{\pi}{2}\right) = 3 - \cos\frac{\pi}{2} = 3 - 0 = 3$.

2a Taking the primitive,
$f(x) = \frac{1}{2}\sin 2x + C$.
Substituting $x = \frac{\pi}{4}$, $1 = \frac{1}{2}\sin\frac{\pi}{2} + C$,
so $1 = \frac{1}{2} + C$ and $C = \frac{1}{2}$.
Hence $f(x) = \frac{1}{2}\sin 2x + \frac{1}{2}$.

2b
$f\left(\frac{\pi}{12}\right) = \frac{1}{2}\sin\frac{\pi}{6} + \frac{1}{2} = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} = \frac{3}{4}$

Your turn.

1 Taking the primitive, $y = \cos x + C$.
When $x = 0$, $y = 4$, so
$4 = \cos 0 + C$,
so $4 = 1 + C$ and $C = 3$.
Hence $f(x) = \cos x + 3$.
Then
$f(\pi) = \cos\pi + 3 = -1 + 3 = 2$.

2a Taking the primitive,
$f(x) = \frac{1}{2}\sin 2x + C$.
Substituting $x = \frac{\pi}{2}$, $2 = \frac{1}{2}\sin\pi + C$,
so $2 = 0 + C$ and $C = 2$.
Hence $f(x) = \frac{1}{2}\sin 2x + 2$.

2b
$f\left(\frac{\pi}{4}\right) = \frac{1}{2}\sin\frac{\pi}{2} + 2 = \frac{1}{2} \times 1 + 2 = 2\frac{1}{2}$

C6 Tutor demo.

a Let $y = \cos^7 x$ and $u = \cos x$. Then $y = u^7$, $\frac{du}{dx} = -\sin x$ and $\frac{dy}{du} = 7u^6$. By the chain rule, $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = 7u^6 \times (-\sin x) = -7\sin x \cos^6 x$.

b Reversing part **a**, $\int (-7\sin x \cos^6 x)\,dx = \cos^7 x$, and dividing by -7,
$\int \sin x \cos^6 x\,dx = -\frac{1}{7}\cos^7 x$. Hence
$\int_0^{\pi} \sin x \cos^6 x\,dx = -\frac{1}{7}\left[\cos^7 x\right]_0^{\pi} = -\frac{1}{7}(-1 - 1) = \frac{2}{7}$,
because $\cos^7\pi = (-1)^7 = -1$ and $\cos^7 0 = 1$.

Your turn.

a Let $y = \cos^3 x$ and $u = \cos x$. Then $y = u^3$, $\frac{du}{dx} = -\sin x$ and $\frac{dy}{du} = 3u^2$. By the chain rule, $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = 3u^2 \times (-\sin x) = -3\sin x \cos^2 x$.

b Reversing part **a**, $\int (-3\sin x \cos^2 x)\,dx = \cos^3 x$, and dividing by -3,
$\int \sin x \cos^2 x\,dx = -\frac{1}{3}\cos^3 x$. Hence
$\int_0^{\frac{\pi}{3}} \sin x \cos^2 x\,dx = -\frac{1}{3}\left[\cos^3 x\right]_0^{\frac{\pi}{3}} = -\frac{1}{3}\left(\frac{1}{8} - 1\right) = \frac{7}{24}$,

because $\cos^3 \frac{\pi}{3} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ and $\cos^3 0 = 1$.

C7 **Tutor demo.** Let $u = \sin x$. Then $u' = \cos x$, and with $n = 6$, $\int u^6 \dfrac{du}{dx}\, dx = \dfrac{u^7}{7}$.

$$\begin{aligned}\int \cos x \sin^6 x\, dx &= \int (\sin x)^6 \cos x\, dx \\ &= \tfrac{1}{7}\sin^7 x + C\end{aligned}$$

Your turn. Let $u = \cos x$. Then $u' = -\sin x$, and with $n = 6$, $\int u^6 \dfrac{du}{dx}\, dx = \dfrac{u^7}{7}$.

$$\begin{aligned}\int \sin x \cos^6 x\, dx &= -\int (-\sin x) \cos^6 x\, dx \\ &= -\tfrac{1}{7}\cos^7 x + C\end{aligned}$$

Textbook Homework

1 **a** $\tan x + C$ **b** $\sin x + C$ **c** $-\cos x + C$ **d** $\cos x + C$ **e** $2\sin x + C$
f $\frac{1}{2}\sin 2x + C$ **g** $\frac{1}{2}\sin x + C$ **h** $2\sin \frac{1}{2}x + C$ **i** $-\frac{1}{2}\cos 2x + C$
j $\frac{1}{5}\tan 5x + C$ **k** $\frac{1}{3}\sin 3x + C$ **l** $3\tan \frac{1}{3}x + C$ **m** $-2\cos \frac{x}{2} + C$
n $-5\sin \frac{1}{5}x + C$ **o** $2\cos 2x + C$ **p** $-\cos \frac{1}{4}x + C$ **q** $36\tan \frac{1}{3}x + C$
r $6\sin \frac{x}{3} + C$

2 **a** 1 **b** $\frac{1}{2}$ **c** $\frac{1}{\sqrt{2}}$ **d** $\sqrt{3}$ **e** 1 **f** $\frac{3}{4}$ **g** 2 **h** 1 **i** 4

3 **a** $y = 1 - \cos x$ **b** $y = \sin x + \cos 2x - 1$ **c** $y = -\cos x + \sin x - 3$

4 **a** about 200 little squares, that is, about 2 square units
b about 29, 100, 171, 171, 100 and 29 little squares: **i** $\doteq 0.29$ **ii** 1 **iii** $\doteq 1.71$ **iv** $\doteq 1.71$ **v** 1 **vi** $\doteq 0.29$
c i $1 - \frac{1}{\sqrt{2}}$ **ii** 1 **iii** $1 + \frac{1}{\sqrt{2}}$ **iv** $1 + \frac{1}{\sqrt{2}}$ **v** 1 **vi** $1 - \frac{1}{\sqrt{2}}$

6 **a** $\sin(x+2) + C$ **b** $\frac{1}{2}\sin(2x+1) + C$ **c** $-\cos(x+2) + C$
d $-\frac{1}{2}\cos(2x+1) + C$ **e** $\frac{1}{3}\sin(3x-2) + C$ **f** $\frac{1}{5}\cos(7-5x) + C$
g $-\tan(4-x) + C$ **h** $-3\tan\left(\dfrac{1-x}{3}\right) + C$ **i** $3\cos\left(\dfrac{1-x}{3}\right) + C$

7 **a** $2\sin 3x + 8\cos \frac{1}{2}x + C$ **b** $4\tan 2x - 40\sin \frac{1}{4}x - 36\cos \frac{1}{3}x + C$

8 **a** $f(x) = \sin \pi x$, $f\left(\frac{1}{3}\right) = \frac{1}{2}\sqrt{3}$ **b** $f(x) = \dfrac{1}{2\pi} + \dfrac{1}{\pi}\sin \pi x$, $f\left(\frac{1}{6}\right) = \dfrac{1}{\pi}$
c $f(x) = -2\cos 3x + x + \left(1 - \frac{\pi}{2}\right)$

9 **a** $-\cos(ax+b) + C$ **b** $\pi \sin \pi x + C$ **c** $\dfrac{1}{u^2}\tan(v+ux) + C$
d $\tan ax + C$

10 a $1+\tan^2 x=\sec^2 x,\ \tan x-x+C$ b $1-\sin^2 x=\cos^2 x,\ 2\sqrt{3}$

11 a $\log_e|f(x)|+C$

12 a $\displaystyle\int \tan x\,dx=-\ln|\cos x|+C$ b $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{2}}\cot x\,dx=\Big[\ln|\sin x|\Big]_{\frac{\pi}{6}}^{\frac{\pi}{2}}=\ln 2$

13 a $5\sin^4 x\cos x,\ \frac{1}{5}\sin^5 x+C$ b $3\tan^2 x\sec^2 x,\ \frac{1}{3}\tan^3 x+C$

14 a $\cos x\,e^{\sin x},\ e-1$ b $\sec^2 x\,e^{\tan x},\ e-1$

15 a 1 b $\frac{5}{24}$

16 a $\frac{2}{9}$ b $\dfrac{1}{n+1}$ c 0 d $\dfrac{1}{2^{n+1}(n+1)}$ e $10\frac{1}{8}$ f $\dfrac{1}{n+1}$

17 b $\sin^2 x=\frac{1}{2}(1-\cos 2x)$, so
$\frac{1}{2}\sin^2 x+C=\frac{1}{4}-\frac{1}{4}\cos 2x+C=-\frac{1}{4}\cos 2x+\left(C+\frac{1}{4}\right)=-\frac{1}{4}\cos 2x+D$,
where $D=C+\frac{1}{4}$.

18 $\sin 2x+2x\cos 2x,\ \dfrac{\pi-2}{8}$

19 b $\frac{4}{3}$

20 b i $\frac{6}{5}$ ii $-\frac{6}{7}$

21 a $A=5,\ B=3$

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 7D | Integrating the trigonometric functions

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q7**

Write $\tan x = \dfrac{\sin x}{\cos x}$. With $u = \cos x$, $u' = -\sin x$, so the top becomes the derivative of the bottom after balancing with -1:

$$\begin{aligned}\int \tan x \, dx &= -\int \frac{-\sin x}{\cos x}\, dx \\ &= -\ln|\cos x| + c.\end{aligned}$$

Write the constant as $c = -\ln A$, where $A > 0$. Then
$-\ln|\cos x| - \ln A = -\big(\ln|\cos x| + \ln A\big) = -\ln|A\cos x|$.

The answer is **(D)**. Option (A) is wrong because $\sec^2 x$ is the derivative of $\tan x$, not a primitive. Option (B) is a primitive of $\cot x$. Option (C) is the derivative of $\tan^{-1} x$, and has nothing to do with $\tan x$.

PP 2 **Caringbah High School, 2021, Year 12 Mathematics Extension 1 HSC Course Assessment Task 4, Q6**

Let $u = \tan\theta$. Then $\dfrac{du}{d\theta} = \sec^2\theta$, and with $n = 2$, $\displaystyle\int u^2 \frac{du}{d\theta}\, d\theta = \frac{u^3}{3}$. So

$$\begin{aligned}\int \sec^2\theta \tan^2\theta \, d\theta &= \int (\tan\theta)^2 \sec^2\theta \, d\theta \\ &= \tfrac{1}{3}\tan^3\theta + C.\end{aligned}$$

Check: $\dfrac{d}{d\theta}\left(\tfrac{1}{3}\tan^3\theta\right) = \tfrac{1}{3} \times 3\tan^2\theta \times \sec^2\theta = \sec^2\theta\tan^2\theta$. The answer is **(B)**. Differentiating (C) gives the extra term $\tan^4\theta\sec^2\theta$, and (A) and (D) do not differentiate back to the integrand either.

PP 3 **Cherrybrook Technology High School, 2020, Year 12 AP4, Mathematics Extension 1, Q5**

Let $u = \cos x$. Then $u' = -\sin x$, and with $n = 3$, $\displaystyle\int u^3 \frac{du}{dx}\,dx = \frac{u^4}{4}$. So

$$\begin{aligned} y &= \int \sin x \cos^3 x \, dx \\ &= -\int (-\sin x)\cos^3 x \, dx \\ &= -\tfrac{1}{4}\cos^4 x + C. \end{aligned}$$

Check: $\dfrac{d}{dx}\left(-\tfrac{1}{4}\cos^4 x\right) = -\tfrac{1}{4} \times 4\cos^3 x \times (-\sin x) = \sin x \cos^3 x$. The answer is **B**. Option A has the wrong sign, and C and D do not differentiate back to $\sin x \cos^3 x$.

PP 4 **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(b)**

Let $u = \sin x$. Then $u' = \cos x$, and with $n = 1$, $\displaystyle\int u \frac{du}{dx}\,dx = \frac{u^2}{2}$. So

$$\begin{aligned} \int_0^{\frac{\pi}{6}} \sin x \cos x \, dx &= \left[\tfrac{1}{2}\sin^2 x\right]_0^{\frac{\pi}{6}} \\ &= \tfrac{1}{2}\sin^2 \tfrac{\pi}{6} - \tfrac{1}{2}\sin^2 0 \\ &= \tfrac{1}{2} \times \left(\tfrac{1}{2}\right)^2 - 0 \\ &= \tfrac{1}{2} \times \tfrac{1}{4} \\ &= \tfrac{1}{8}. \end{aligned}$$

PP 5 **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(e)**

Integrate $y'' = -\sec^2(2x)$, using $\displaystyle\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + C$ with $a = 2$:

$y' = -\frac{1}{2}\tan 2x + C.$

The gradient is -1 at $x = \dfrac{\pi}{8}$, and $2 \times \dfrac{\pi}{8} = \dfrac{\pi}{4}$, so

$-1 = -\frac{1}{2}\tan\frac{\pi}{4} + C$

$-1 = -\frac{1}{2} \times 1 + C$

$C = -\frac{1}{2}.$

Hence $y' = -\frac{1}{2}\tan 2x - \frac{1}{2}$. At $x = \dfrac{\pi}{12}$, $2 \times \dfrac{\pi}{12} = \dfrac{\pi}{6}$, so

$$\begin{aligned} y' &= -\tfrac{1}{2}\tan\tfrac{\pi}{6} - \tfrac{1}{2} \\ &= -\tfrac{1}{2} \times \tfrac{1}{\sqrt{3}} - \tfrac{1}{2} \\ &= -\tfrac{\sqrt{3}}{6} - \tfrac{3}{6} \\ &= -\frac{3+\sqrt{3}}{6}. \end{aligned}$$

The exact gradient at $x = \dfrac{\pi}{12}$ is $-\dfrac{3+\sqrt{3}}{6}$. (The y-value 2 at the point is not needed for the gradient. It would be needed to find y itself.)

PP 6 **Hornsby Girls High School, 2022, Mathematics Extension 1, Year 12 Higher School Certificate Trial Examination Term 3, Q14(a)**

Write $\cot x = \dfrac{\cos x}{\sin x}$. The top is the derivative of the bottom, so

$$\begin{aligned} f(x) &= \int \left(\frac{\cos x}{\sin x} + x \right) dx \\ &= \ln |\sin x| + \frac{x^2}{2} + C. \end{aligned}$$

Substituting $f\left(\dfrac{\pi}{2}\right) = 0$,

$$\begin{aligned} \ln \left| \sin \tfrac{\pi}{2} \right| + \frac{\left(\frac{\pi}{2}\right)^2}{2} + C &= 0 \\ \ln 1 + \frac{\pi^2}{8} + C &= 0 \\ C &= -\frac{\pi^2}{8}, \end{aligned}$$

because $\sin \frac{\pi}{2} = 1$, $\ln 1 = 0$ and $\left(\frac{\pi}{2}\right)^2 \div 2 = \frac{\pi^2}{4} \div 2 = \frac{\pi^2}{8}$. Hence

$$f(x) = \ln |\sin x| + \frac{x^2}{2} - \frac{\pi^2}{8}.$$

PP 7 **Abbotsleigh, 2023, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(d)(ii)**

A primitive of $2\sqrt{3}\cos x - 2\sin x$ is $2\sqrt{3}\sin x + 2\cos x$, so

$$\begin{aligned} \int_0^k \left(2\sqrt{3}\cos x - 2\sin x\right) dx &= \Big[2\sqrt{3}\sin x + 2\cos x\Big]_0^k \\ &= \left(2\sqrt{3}\sin k + 2\cos k\right) - \left(2\sqrt{3}\sin 0 + 2\cos 0\right) \\ &= 2\sqrt{3}\sin k + 2\cos k - 2. \end{aligned}$$

Setting this equal to 2,

$$\begin{aligned} 2\sqrt{3}\sin k + 2\cos k - 2 &= 2 \\ 2\sqrt{3}\sin k + 2\cos k &= 4 \\ 4\sin\left(k + \tfrac{\pi}{6}\right) &= 4 \qquad \text{(using part (i))} \\ \sin\left(k + \tfrac{\pi}{6}\right) &= 1. \end{aligned}$$

Since $0 \le k \le 2\pi$, the angle $k + \frac{\pi}{6}$ lies between $\frac{\pi}{6}$ and $\frac{13\pi}{6}$. In this interval, $\sin\theta = 1$ only at $\theta = \frac{\pi}{2}$ (the next solution, $\frac{5\pi}{2}$, is too large). So

$$k + \tfrac{\pi}{6} = \tfrac{\pi}{2}, \qquad k = \tfrac{\pi}{2} - \tfrac{\pi}{6} = \tfrac{\pi}{3}.$$

PP 8 **Killara High School, 2020, Mathematics Extension 1 Trial HSC Examination, Q12(d)(ii)**

By part (i), $\operatorname{cosec} x + \cot x = \cot \dfrac{x}{2} = \dfrac{\cos \frac{x}{2}}{\sin \frac{x}{2}}$. Let $u = \sin \dfrac{x}{2}$. Then $u' = \dfrac{1}{2}\cos \dfrac{x}{2}$, so balance with 2 outside:

$$\begin{aligned}
\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x)\, dx &= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot \tfrac{x}{2}\, dx \\
&= 2\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\frac{1}{2}\cos \frac{x}{2}}{\sin \frac{x}{2}}\, dx \\
&= 2\left[\ln \left| \sin \tfrac{x}{2} \right| \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} \\
&= 2\left(\ln \sin \tfrac{\pi}{4} - \ln \sin \tfrac{\pi}{6} \right) \\
&= 2\left(\ln \tfrac{1}{\sqrt{2}} - \ln \tfrac{1}{2} \right) \\
&= 2\ln \left(\tfrac{1}{\sqrt{2}} \div \tfrac{1}{2} \right) \\
&= 2\ln \tfrac{2}{\sqrt{2}} \\
&= 2\ln \sqrt{2} \\
&= \ln 2.
\end{aligned}$$

Here $\dfrac{2}{\sqrt{2}} = \sqrt{2}$ and $2\ln\sqrt{2} = \ln(\sqrt{2})^2 = \ln 2$. The interval $\dfrac{\pi}{3} \le x \le \dfrac{\pi}{2}$ does not contain a zero of $\sin \dfrac{x}{2}$, so no asymptote is crossed.

Part 2 Exp, Log and Trig Integrals - Homework Answers

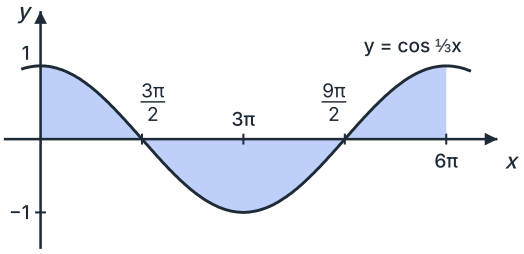
Lesson 7E | Applications of integration

Lesson Examples

C1 **Tutor demo.**

a The curve $y = \cos \frac{1}{3}x$ has amplitude 1, and the period is $2\pi \div \frac{1}{3} = 6\pi$.

It crosses the x-axis where $\frac{1}{3}x = \frac{\pi}{2}$ or $\frac{1}{3}x = \frac{3\pi}{2}$, so the two x-intercepts in the interval are $x = \frac{3\pi}{2}$ and $x = \frac{9\pi}{2}$.



b Integrate separately over $\left[0, \frac{3\pi}{2}\right]$, $\left[\frac{3\pi}{2}, \frac{9\pi}{2}\right]$ and $\left[\frac{9\pi}{2}, 6\pi\right]$. The primitive of $\cos \frac{1}{3}x$ is $\frac{1}{\frac{1}{3}} \sin \frac{1}{3}x = 3 \sin \frac{1}{3}x$.

$$\begin{aligned}
\int_0^{\frac{3\pi}{2}} \cos \tfrac{1}{3}x \, dx &= \left[3 \sin \tfrac{1}{3}x\right]_0^{\frac{3\pi}{2}} \\
&= 3 \sin \tfrac{\pi}{2} - 3 \sin 0 \\
&= 3 - 0 \\
&= 3,
\end{aligned}$$

which is positive, because the curve is above the x-axis there.

$$\begin{aligned}
\int_{\frac{3\pi}{2}}^{\frac{9\pi}{2}} \cos \tfrac{1}{3}x \, dx &= \left[3 \sin \tfrac{1}{3}x\right]_{\frac{3\pi}{2}}^{\frac{9\pi}{2}} \\
&= 3 \sin \tfrac{3\pi}{2} - 3 \sin \tfrac{\pi}{2} \\
&= -3 - 3 \\
&= -6,
\end{aligned}$$

which is negative, because the curve is below the x-axis there.

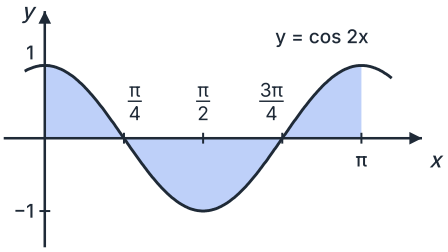
$$\begin{aligned}
\int_{\frac{9\pi}{2}}^{6\pi} \cos \tfrac{1}{3}x \, dx &= \left[3 \sin \tfrac{1}{3}x\right]_{\frac{9\pi}{2}}^{6\pi} \\
&= 3 \sin 2\pi - 3 \sin \tfrac{3\pi}{2} \\
&= 0 - (-3) \\
&= 3,
\end{aligned}$$

which is positive, because the curve is above the x-axis there. At the limits,
$\frac{1}{3} \times \frac{3\pi}{2} = \frac{\pi}{2}$, $\frac{1}{3} \times \frac{9\pi}{2} = \frac{3\pi}{2}$ and $\frac{1}{3} \times 6\pi = 2\pi$.

$$\begin{aligned}\textbf{Hence total area} &= 3 + 6 + 3 \\ &= \textbf{12 square units.}\end{aligned}$$

Your turn.

a The curve $y = \cos 2x$ has amplitude 1, and the period is $2\pi \div 2 = \pi$.
It crosses the x-axis where $2x = \frac{\pi}{2}$ or $2x = \frac{3\pi}{2}$, so the two x-intercepts in the interval are $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$.



b Integrate separately over $\left[0, \frac{\pi}{4}\right]$, $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ and $\left[\frac{3\pi}{4}, \pi\right]$. The primitive of $\cos 2x$ is $\frac{1}{2}\sin 2x$.

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \cos 2x \, dx &= \left[\tfrac{1}{2}\sin 2x\right]_0^{\frac{\pi}{4}} \\ &= \tfrac{1}{2}\sin\tfrac{\pi}{2} - \tfrac{1}{2}\sin 0 \\ &= \tfrac{1}{2} - 0 \\ &= \tfrac{1}{2},\end{aligned}$$

which is positive, because the curve is above the x-axis there.

$$\begin{aligned}\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \cos 2x \, dx &= \left[\tfrac{1}{2}\sin 2x\right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \\ &= \tfrac{1}{2}\sin\tfrac{3\pi}{2} - \tfrac{1}{2}\sin\tfrac{\pi}{2} \\ &= -\tfrac{1}{2} - \tfrac{1}{2} \\ &= -1,\end{aligned}$$

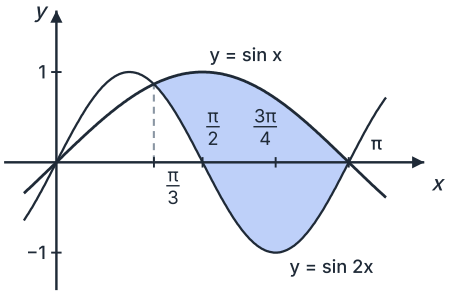
which is negative, because the curve is below the x-axis there.

$$\begin{aligned}\int_{\frac{3\pi}{4}}^{\pi} \cos 2x \, dx &= \left[\tfrac{1}{2}\sin 2x\right]_{\frac{3\pi}{4}}^{\pi} \\ &= \tfrac{1}{2}\sin 2\pi - \tfrac{1}{2}\sin\tfrac{3\pi}{2} \\ &= 0 - \left(-\tfrac{1}{2}\right) \\ &= \tfrac{1}{2},\end{aligned}$$

which is positive, because the curve is above the x-axis there.

$$\begin{aligned}\textbf{Hence total area} &= \tfrac{1}{2} + 1 + \tfrac{1}{2} \\ &= \textbf{2 square units.}\end{aligned}$$

C2 **Tutor demo.**



a At $x = \dfrac{\pi}{3}$, $\sin x = \sin\dfrac{\pi}{3} = \frac{1}{2}\sqrt{3}$ and $\sin 2x = \sin\dfrac{2\pi}{3} = \frac{1}{2}\sqrt{3}$. The y-values are equal, so the curves intersect when $x = \dfrac{\pi}{3}$.

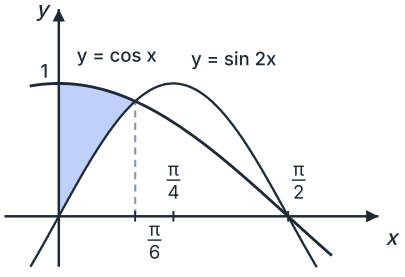
b The curves are sketched to the right. They also meet at $x = 0$ and $x = \pi$.

c In the interval $\dfrac{\pi}{3} \le x \le \pi$, the curve $y = \sin x$ is above $y = \sin 2x$: at $x = \dfrac{\pi}{2}$, $\sin x = 1$ and $\sin 2x = \sin\pi = 0$.

$$\begin{aligned}\textbf{area between} &= \int_{\frac{\pi}{3}}^{\pi} (\sin x - \sin 2x)\, dx \\ &= \left[-\cos x + \tfrac{1}{2}\cos 2x \right]_{\frac{\pi}{3}}^{\pi} \\ &= \left(-\cos\pi + \tfrac{1}{2}\cos 2\pi \right) - \left(-\cos\tfrac{\pi}{3} + \tfrac{1}{2}\cos\tfrac{2\pi}{3} \right) \\ &= \left(1 + \tfrac{1}{2}\right) - \left(-\tfrac{1}{2} - \tfrac{1}{4} \right) \\ &= \tfrac{3}{2} + \tfrac{3}{4} \\ &= \tfrac{9}{4}\ \textbf{square units.}\end{aligned}$$

Here $\cos\pi = -1$, $\cos 2\pi = 1$, $\cos\dfrac{\pi}{3} = \frac{1}{2}$ and $\cos\dfrac{2\pi}{3} = -\frac{1}{2}$, so $\frac{1}{2}\cos\dfrac{2\pi}{3} = -\frac{1}{4}$.

Your turn.



a At $x = \dfrac{\pi}{6}$, $\cos x = \cos\dfrac{\pi}{6} = \frac{1}{2}\sqrt{3}$ and $\sin 2x = \sin\dfrac{\pi}{3} = \frac{1}{2}\sqrt{3}$. The y-values are equal, so the curves intersect when $x = \dfrac{\pi}{6}$.

b The curves are sketched to the right. They also meet at $x = \dfrac{\pi}{2}$.

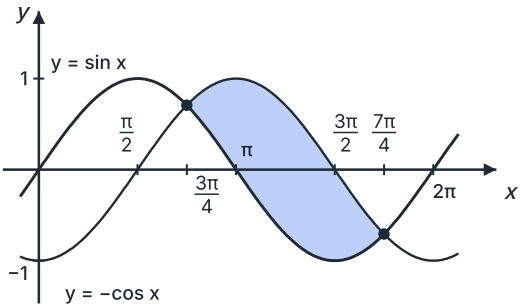
c In the interval $0 \le x \le \dfrac{\pi}{6}$, the curve $y = \cos x$ is above $y = \sin 2x$: at $x = 0$, $\cos x = 1$ and $\sin 2x = 0$.

$$\begin{aligned}\text{area between} &= \int_0^{\frac{\pi}{6}} (\cos x - \sin 2x)\,dx \\ &= \left[\sin x + \tfrac{1}{2}\cos 2x\right]_0^{\frac{\pi}{6}} \\ &= \left(\sin\tfrac{\pi}{6} + \tfrac{1}{2}\cos\tfrac{\pi}{3}\right) - \left(\sin 0 + \tfrac{1}{2}\cos 0\right) \\ &= \left(\tfrac{1}{2} + \tfrac{1}{4}\right) - \left(0 + \tfrac{1}{2}\right) \\ &= \tfrac{1}{4} \text{ square units.}\end{aligned}$$

The primitive of $-\sin 2x$ is $\frac{1}{2}\cos 2x$. At the upper limit, $2 \times \frac{\pi}{6} = \frac{\pi}{3}$.

C3 Tutor demo.

a Put $\sin x = -\cos x$. Then $\tan x = -1$, so $x = \dfrac{3\pi}{4}$ or $\dfrac{7\pi}{4}$ (the tangent is negative in quadrants 2 and 4, and the related angle is $\dfrac{\pi}{4}$). So the curves intersect at the points $\left(\dfrac{3\pi}{4}, \tfrac{1}{2}\sqrt{2}\right)$ and $\left(\dfrac{7\pi}{4}, -\tfrac{1}{2}\sqrt{2}\right)$.



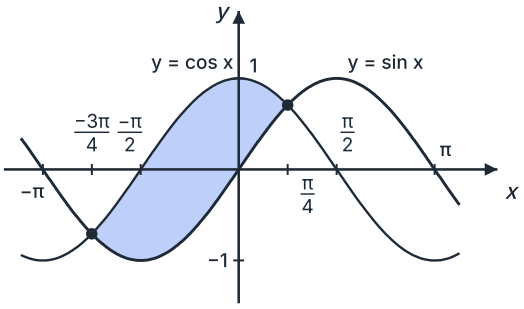
b Between the intersections, $y = -\cos x$ is above $y = \sin x$: at $x = \pi$, $-\cos\pi = 1$ and $\sin\pi = 0$.

$$\begin{aligned}\text{Area between} &= \int_{\frac{3\pi}{4}}^{\frac{7\pi}{4}} (-\cos x - \sin x)\,dx \\ &= \left[-\sin x + \cos x\right]_{\frac{3\pi}{4}}^{\frac{7\pi}{4}} \\ &= \left(-\sin\tfrac{7\pi}{4} + \cos\tfrac{7\pi}{4}\right) - \left(-\sin\tfrac{3\pi}{4} + \cos\tfrac{3\pi}{4}\right) \\ &= \left(\tfrac{1}{2}\sqrt{2} + \tfrac{1}{2}\sqrt{2}\right) - \left(-\tfrac{1}{2}\sqrt{2} - \tfrac{1}{2}\sqrt{2}\right) \\ &= \sqrt{2} + \sqrt{2} \\ &= 2\sqrt{2} \text{ square units.}\end{aligned}$$

Here $\sin\dfrac{7\pi}{4} = -\tfrac{1}{2}\sqrt{2}$, $\cos\dfrac{7\pi}{4} = \tfrac{1}{2}\sqrt{2}$, $\sin\dfrac{3\pi}{4} = \tfrac{1}{2}\sqrt{2}$ and $\cos\dfrac{3\pi}{4} = -\tfrac{1}{2}\sqrt{2}$.

Your turn.

a Put $\sin x = \cos x$. Then $\tan x = 1$, so $x = -\dfrac{3\pi}{4}$ or $\dfrac{\pi}{4}$ (the tangent is positive in quadrants 1 and 3, and the related angle is $\dfrac{\pi}{4}$). So the curves intersect at the points $\left(-\dfrac{3\pi}{4}, -\frac{1}{2}\sqrt{2}\right)$ and $\left(\dfrac{\pi}{4}, \frac{1}{2}\sqrt{2}\right)$.



b Between the intersections, $y = \cos x$ is above $y = \sin x$: at $x = 0$, $\cos 0 = 1$ and $\sin 0 = 0$.

$$\begin{aligned}
\text{Area between} &= \int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}} (\cos x - \sin x)\, dx \\
&= \Big[\sin x + \cos x\Big]_{-\frac{3\pi}{4}}^{\frac{\pi}{4}} \\
&= \left(\sin\tfrac{\pi}{4} + \cos\tfrac{\pi}{4}\right) - \left(\sin\left(-\tfrac{3\pi}{4}\right) + \cos\left(-\tfrac{3\pi}{4}\right)\right) \\
&= \left(\tfrac{1}{2}\sqrt{2} + \tfrac{1}{2}\sqrt{2}\right) - \left(-\tfrac{1}{2}\sqrt{2} - \tfrac{1}{2}\sqrt{2}\right) \\
&= \sqrt{2} + \sqrt{2} \\
&= 2\sqrt{2} \text{ square units.}
\end{aligned}$$

Textbook Homework

1 **a** 1 square unit **b** $\frac{1}{2}$ square unit

2 **a** 1 square unit **b** $\sqrt{3}$ square units

3 **a** $1 - \dfrac{1}{\sqrt{2}}$ square units **b** $1 - \dfrac{\sqrt{3}}{2}$ square units

4 **a** $\frac{1}{2}\sqrt{3}\ \text{u}^2$ **b** $\frac{1}{2}\sqrt{3}\ \text{u}^2$

5 **a** $\frac{1}{2}\ \text{u}^2$ **b** $\frac{1}{2}\ \text{u}^2$ **c** $1 - \dfrac{\sqrt{3}}{2} = \frac{1}{2}\left(2 - \sqrt{3}\right)\ \text{u}^2$
d $\frac{1}{3}\left(1 - \dfrac{1}{\sqrt{2}}\right) = \frac{1}{6}\left(2 - \sqrt{2}\right)\ \text{u}^2$ **e** $\frac{2}{3}\sqrt{3}\ \text{u}^2$ **f** $4\ \text{u}^2$

6 **a** $\left(\sqrt{2}-1\right)$ u^2 **b** $\frac{1}{4}$ u^2 **c** $\left(\frac{\pi^2}{8}-1\right)$ u^2 **d** $(\pi-2)$ u^2

7 **a** $\left(2-\sqrt{2}\right)$ u^2 **b** $1\frac{1}{2}$ u^2

8 **a** 2 u^2 **b** 1 u^2

9 **a** 2 u^2 **b** $\sqrt{2}$ u^2 **c** 2 u^2 **d** $\frac{1}{2}$ u^2 **e** 4 u^2 **f** 1 u^2

10 **b** $\frac{4}{\pi}$ u^2

11 3.8 m^2

12 4 u^2

14 **b** $\frac{1}{2}\left(3+\sqrt{3}\right)$ u^2

15 **c** $\frac{3}{4}\sqrt{3}$ u^2

16 **b** $2\sqrt{2}$ u^2

17 **b** They are all 4 u^2.

18 **b** The curve is below $y=1$ just as much as it is above $y=1$, so the area is equal to the area of a rectangle n units long and 1 unit high.

19 12

20 **a** Since $x^2>0$ for $0<x<\frac{\pi}{2}$, $x^2\sin x<x^2\times x<x^2\tan x$.

21 **b** $\cos x$ and $(1+\sin x)^2$ are both positive in the given domain, so y' is negative there.

22 **a** 0, since the integrand is odd. **b** 0, since the integrand is odd.
c 2, since the integrand is even. **d** $6\sqrt{3}$, since the integrand is even.
e 6π. The first term is even, the other two are odd.
f $-\frac{2}{3}+\frac{\pi^3}{4}$. The first term is odd, the other two are even.

23 **b** $\int_0^N e^{-x}\sin x\,dx=\frac{1}{2}-\frac{1}{2}e^{-N}(\sin N+\cos N)$, which tends to $\frac{1}{2}$ as $N\to\infty$.
c $\int_0^{\pi} e^{-x}\sin x\,dx=\frac{1}{2}\left(1+e^{-\pi}\right)$, $\int_{2\pi}^{3\pi} e^{-x}\sin x\,dx=\frac{1}{2}e^{-2\pi}\left(1+e^{-\pi}\right)$, so the ratio is $e^{-2\pi}$.

Part 2 Exp, Log and Trig Integrals - Homework Answers

Lesson 7E | Applications of integration

Past-Paper Homework - Worked Solutions

PP 1 **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q10**

The points A and B are where the curves meet. Put $\sin x = \cos x$. Then $\tan x = 1$, so $x = \frac{\pi}{4}$ or $x = \frac{5\pi}{4}$. These are the limits of the integral.

Between A and B, the curve $y = \sin x$ is on top: at $x = \frac{\pi}{2}$, $\sin x = 1$ and $\cos x = 0$. The area is the integral of the top curve minus the bottom curve, $\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \left(\sin(x) - \cos(x)\right) dx$.

Option A is the bottom curve minus the top curve, so it gives the opposite of the area. Option C adds the two functions instead of subtracting them. Option D has the wrong limits. **Answer (B).**

PP 2 **Caringbah High School, 2021, Year 12 Mathematics Extension 1 HSC Course Assessment Task 4, Q12(c)**

(i) When $x = 0$,

$$\begin{aligned} y &= 2\cos^{-1} 0 \\ &= 2 \times \frac{\pi}{2} \\ &= \pi. \end{aligned}$$

The y intercept is π.

(ii) The domain of $\cos^{-1}$ is $-1 \leq \frac{x}{3} \leq 1$. Multiplying by 3, the domain of f is $-3 \leq x \leq 3$.

The range of $\cos^{-1}$ is $0 \leq \cos^{-1} \frac{x}{3} \leq \pi$. Multiplying by 2, the range of f is $0 \leq y \leq 2\pi$.

(iii) The region lies under the curve between $x = 0$ and $x = 3$. At $x = 3$, $y = 2\cos^{-1} 1 = 0$, and at $x = 0$, $y = \pi$. So the region is bounded by the curve, the x-axis and the y-axis, and it runs from $y = 0$ to $y = \pi$. Make x the subject, and integrate with respect to y:

$$\begin{aligned} y &= 2\cos^{-1} \frac{x}{3} \\ \frac{y}{2} &= \cos^{-1} \frac{x}{3} \\ \frac{x}{3} &= \cos \frac{y}{2} \\ x &= 3\cos \frac{y}{2}. \end{aligned}$$

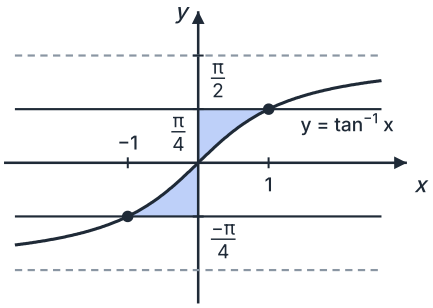
For $0 \leq y \leq \pi$, $x = 3\cos\dfrac{y}{2}$ is positive, so the region lies between the y-axis and this curve.

$$\begin{aligned}\text{Area} &= \int_0^{\pi} 3\cos\frac{y}{2}\,dy \\ &= \left[6\sin\frac{y}{2}\right]_0^{\pi} \\ &= 6\sin\frac{\pi}{2} - 6\sin 0 \\ &= 6\times 1 - 0 \\ &= 6.\end{aligned}$$

The area is 6 square units. (The primitive of $\cos\frac{1}{2}y$ is $\dfrac{1}{\frac{1}{2}}\sin\frac{1}{2}y = 2\sin\frac{1}{2}y$, and $3\times 2 = 6$.)

PP 3 **Northern Beaches Secondary College Manly Selective Campus, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q11(c)**

The curve $y = \tan^{-1}x$ passes through the origin and meets the lines $y = \pm\dfrac{\pi}{4}$ at $x = \pm 1$, because $\tan^{-1}1 = \dfrac{\pi}{4}$ and $\tan^{-1}(-1) = -\dfrac{\pi}{4}$. The region has two parts, one in the first quadrant and one in the third quadrant, each between the curve and the y-axis.



Make x the subject: from $y = \tan^{-1}x$, $x = \tan y$, for $-\dfrac{\pi}{2} < y < \dfrac{\pi}{2}$.

The function $\tan^{-1}x$ is odd, so the two parts have equal area. In the first quadrant, the region runs from $y = 0$ to $y = \dfrac{\pi}{4}$, between the y-axis and $x = \tan y$. Using $\displaystyle\int \tan y\,dy = -\log_e|\cos y| + C$ (Lesson 7D),

$$\begin{aligned}\text{Area} &= 2\int_0^{\frac{\pi}{4}} \tan y\,dy \\ &= 2\Big[-\log_e \cos y\Big]_0^{\frac{\pi}{4}} \\ &= 2\Big(-\log_e \cos\tfrac{\pi}{4} + \log_e \cos 0\Big) \\ &= 2\Big(-\log_e \frac{1}{\sqrt{2}} + \log_e 1\Big) \\ &= 2\big(\log_e \sqrt{2} + 0\big) \\ &= 2 \times \tfrac{1}{2}\log_e 2 \\ &= \log_e 2.\end{aligned}$$

The area is $\ln 2$ square units. Here $\cos y > 0$ for $0 \le y \le \dfrac{\pi}{4}$, so $|\cos y| = \cos y$; also $-\log_e \dfrac{1}{\sqrt{2}} = \log_e \sqrt{2}$ and $\log_e \sqrt{2} = \log_e 2^{\frac{1}{2}} = \tfrac{1}{2}\log_e 2$.

PP 4 **Hornsby Girls High School, 2023, Year 12 Higher School Certificate Trial Examination Term 3, Mathematics Extension 1, Q14(b)**

The shaded region is bounded by the curve and the two axes.

When $x = 0$, $y = 2\cos^{-1}\tfrac{1}{2} = 2 \times \dfrac{\pi}{3} = \dfrac{2\pi}{3}$.

When $y = 0$, $\cos^{-1}\left(x + \tfrac{1}{2}\right) = 0$, so $x + \tfrac{1}{2} = 1$ and $x = \tfrac{1}{2}$.

So the region runs from $y = 0$ to $y = \dfrac{2\pi}{3}$. Make x the subject:

$$\begin{aligned} y &= 2\cos^{-1}\left(x + \tfrac{1}{2}\right) \\ \frac{y}{2} &= \cos^{-1}\left(x + \tfrac{1}{2}\right) \\ \cos\frac{y}{2} &= x + \tfrac{1}{2} \\ x &= \cos\frac{y}{2} - \tfrac{1}{2}.\end{aligned}$$

The region lies between the y-axis and this curve, so integrate with respect to y:

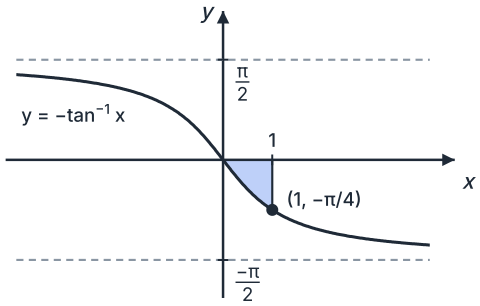
$$
\begin{aligned}
A &= \int_0^{\frac{2\pi}{3}} \left(\cos\frac{y}{2} - \tfrac{1}{2}\right) dy \\
&= \left[2\sin\frac{y}{2} - \tfrac{1}{2}y\right]_0^{\frac{2\pi}{3}} \\
&= \left(2\sin\frac{\pi}{3} - \tfrac{1}{2}\times\frac{2\pi}{3}\right) - \left(2\sin 0 - 0\right) \\
&= 2\times\frac{\sqrt{3}}{2} - \frac{\pi}{3} \\
&= \sqrt{3} - \frac{\pi}{3}.
\end{aligned}
$$

The shaded area is $\left(\sqrt{3} - \dfrac{\pi}{3}\right)$ square units. (At the upper limit, $\frac{1}{2}\times\dfrac{2\pi}{3} = \dfrac{\pi}{3}$.)

PP 5 **Pymble Ladies' College, 2024, Mathematics Extension 1 HSC Trial Examination Term 3, Q13(d)**

(i) The curve $y = -\tan^{-1}x$ is the reflection of $y = \tan^{-1}x$ in the x-axis. It passes through the origin, it is decreasing, and its horizontal asymptotes are $y = \dfrac{\pi}{2}$ (to the left) and $y = -\dfrac{\pi}{2}$ (to the right).

When $x = 1$, $y = -\tan^{-1}1 = -\dfrac{\pi}{4}$, so the labelled point is $\left(1, -\dfrac{\pi}{4}\right)$.



(ii) The region is below the x-axis, from $x = 0$ to $x = 1$, and it runs from $y = -\dfrac{\pi}{4}$ to $y = 0$. Make x the subject: from $y = -\tan^{-1}x$, $\tan^{-1}x = -y$, so $x = \tan(-y) = -\tan y$. At each height y, the region runs from the curve $x = -\tan y$ on the left to the line $x = 1$ on the right. Using the right-hand boundary minus the left-hand boundary,

$$
\begin{aligned}
\text{Area} &= \int_{-\frac{\pi}{4}}^{0} \left(1 - (-\tan y)\right) dy \\
&= \int_{-\frac{\pi}{4}}^{0} \left(1 + \tan y\right) dy \\
&= \Big[y - \log_e \cos y\Big]_{-\frac{\pi}{4}}^{0} \\
&= \left(0 - \log_e \cos 0\right) - \left(-\tfrac{\pi}{4} - \log_e \cos\left(-\tfrac{\pi}{4}\right)\right) \\
&= (0 - 0) - \left(-\tfrac{\pi}{4} - \log_e \frac{1}{\sqrt{2}}\right) \\
&= \frac{\pi}{4} + \log_e \frac{1}{\sqrt{2}} \\
&= \frac{\pi}{4} - \tfrac{1}{2}\log_e 2.
\end{aligned}
$$

The area is $\left(\frac{\pi}{4} - \frac{1}{2}\ln 2\right)$ square units. Here $\cos y > 0$ for $-\frac{\pi}{4} \leq y \leq 0$, so $|\cos y| = \cos y$, and $\log_e \frac{1}{\sqrt{2}} = \log_e 2^{-\frac{1}{2}} = -\frac{1}{2}\log_e 2$.

Part 2 Exp, Log and Trig Integrals - Homework Answers

Review | Chapter review exercise

Chapter 6 review: integration questions

16 a $\frac{1}{5}e^{5x}+C$ b $-2e^{2-5x}+C$ c $5e^{\frac{1}{5}x}+C$ d $\frac{3}{5}e^{5x-4}+C$

17 a e^2-1 b $\frac{1}{2}(e^2-1)$ c $e-1$ d $\frac{1}{3}(e^2-1)$ e $\frac{1}{2}e^2(e-1)$ f $4(e-1)$

18 a $-\frac{1}{5}e^{-5x}+C$ b $\frac{1}{4}e^{4x}+C$ c $-2e^{-3x}+C$ d $\frac{1}{6}e^{6x}+C$
e $-\frac{1}{2}e^{-2x}+C$ f $e^x-\frac{1}{2}e^{-2x}+C$ g $\frac{1}{3}e^{3x}+e^x+C$
h $x-2e^{-x}-\frac{1}{2}e^{-2x}+C$

19 a $2-e^{-1}$ b $\frac{1}{2}(e^4+3)$ c $2(1-e^{-1})$ d $\frac{1}{3}(e-2)$ e $e-e^{-1}$
f $\frac{1}{2}(e^2+4e-3)$

20 $f(x)=e^x+e^{-x}-x+1,\ f(1)=e+e^{-1}$

21 a $3x^2e^{x^3}$ b $\frac{1}{3}(e-1)$

22 a 3.19 u^2 b 0.368 u^2

23 a $\frac{1}{2}(1+e^{-2})$ u^2 b $\frac{1}{2}(3-e)$ u^2

32 a $\log_e|x|+C$ b $3\log_e|x|+C$ c $\frac{1}{5}\log_e|x|+C$ d $\log_e|x+7|+C$
e $\frac{1}{2}\log_e|2x-1|+C$ f $-\frac{1}{3}\log_e|2-3x|+C$ g $\log_e|2x+9|+C$
h $-2\log_e|1-4x|+C$

33 a $\log_e\frac{3}{2}$ b $\frac{1}{4}\log_e 13$ c 1 d 1

34 a $\log_e(x^2+4)+C$ b $\log_e|x^3-5x+7|+C$ c $\frac{1}{2}\log_e|x^2-3|+C$
d $\frac{1}{4}\log_e|x^4-4x|+C$

35 $\log_e 2$ u^2

36 b $(12-5\log_e 5)$ u^2

38 a e^x+C b $\dfrac{2^x}{\log_e 2}+C$ c $\dfrac{3^x}{\log_e 3}+C$ d $\dfrac{5^x}{\log_e 5}+C$

39 a $\dfrac{d}{dx}(x\log_e x)=\log_e x+1,\ \displaystyle\int\log_e x\,dx=x\log_e x-x+C$
b $\dfrac{d}{dx}(xe^x)=e^x+xe^x,\ \displaystyle\int xe^x\,dx=xe^x-e^x+C$

41 **a** $\dfrac{7}{\ln 2}$ and $\dfrac{7}{8\ln 2}$

b When $y = 2^x$ is transformed successively by a vertical dilation with factor 8 and a shift right 3 units, the result is the same graph $y = 2^x$. The region in the second integral is transformed to the region in the first integral by this compound transformation.

Chapter 7 review: integration questions

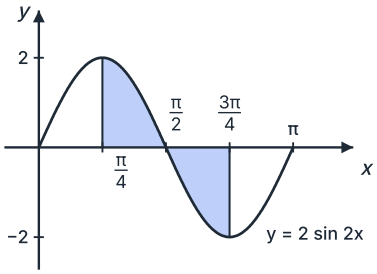
5 **a** $4\sin x + C$ **b** $-\frac{1}{4}\cos 4x + C$ **c** $4\tan \frac{1}{4}x + C$

6 **a** $\sqrt{3} - 1$ **b** $\frac{1}{2}$ **c** $\frac{1}{2}$

7 0.089

8 $y = 2\sin \frac{1}{2}x - 1$

9 **a** **b** 2 u^2



10 **a** $\frac{1}{2}$ u^2 **b** $\dfrac{3\sqrt{3}}{4}$ u^2

11 **a** $\tan x = \dfrac{\sin x}{\cos x}$ **b** $\frac{1}{2}\ln 2$ u^2

16 **a** $\frac{1}{2}\sin e^{2x} + C$ **b** $\frac{1}{2}\cos e^{-2x} + C$ **c** $\frac{1}{3}\log_e |3\tan x + 1| + C$
d $-\frac{3}{5}\log_e |4 + 5\cos x| + C$ **e** $\tan x - \sin x + C$ **f** $\frac{2}{3}$

17 **b i** $\frac{1}{2}\tan^2 x + \log_e |\cos x| + C$ **ii** $\frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x - \log_e |\cos x| + C$

18 **b** $\left(\dfrac{\pi}{4} - \frac{1}{2}\ln 2\right)$ u^2

19 **b i** $1\frac{1}{5}$ **ii** 0
c i $\dfrac{\sin(m+n)x}{2(m+n)} + \dfrac{\sin(m-n)x}{2(m-n)} + C$ **ii** $\dfrac{\sin(m+n)x}{2(m+n)} + \frac{1}{2}x + C$